AccDSL

Company Histories, Accounting Interpretations, and Scenario Simulation

Product and implementation design

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This document specifies a system that translates company disclosures into traceable evidence, represents several compatible economic and accounting histories, and simulates their future journals and financial statements under explicit economic scenarios.

The core requirements, data contracts, semantics and acceptance cases govern implementation. Proposed libraries can be replaced while preserving those contracts. Sector rule packages require their own accounting review before they can be described as compliant with a reporting standard.

Research materials, source qualifications and review records accompany this document in `docs/design/research` and `docs/design/review`.

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Chapter 1

The product decision

A company's accounts describe the financial consequences of its activities, but they leave much of the activity itself unobserved. Revenue combines many sales, a borrowing balance combines contracts with different terms, and a provision combines estimates about uncertain future payments. Aggregation is necessary for reporting. It also makes the inverse task—recovering economic history from published reports—fundamentally different from entering known transactions into a ledger.

AccDSL will preserve this distinction. Its first product is an evidence-backed representation of a company's historical record. Its second is a set of compatible histories and current economic states. Its third is a collection of conditional simulations, each explaining how a particular history, accounting treatment and business response produces future journals and reports. These products share identifiers and provenance, so an analyst can move from a forecast balance to the assumption or disclosure responsible for it.

The recommended design is a small, typed accounting language over a shared intermediate representation, combined with an evidence database, contract calculators and replaceable inference engines. A language earns its place by making recurring accounting decisions explicit and checkable: which entity and period an amount belongs to, whether it is known or assumed, which rule applies, and how an event changes a balanced ledger. The language does not supply missing facts or determine a company's future behaviour.

1.1 Why journals are necessary but insufficient

Consider a company with a borrowing of 100 currency units and annual interest expense of 5. Assume that the entire principal was outstanding throughout the year, interest was paid at year-end, and there were no fees, defaults, hedges or foreign-exchange effects. Both a fixed-rate loan at 5% and a floating-rate loan whose rate happened to be 5% generate the same entries:

Account	Debit	Credit
Cash, on initial borrowing	100	
Borrowing liability		100
Interest expense, at year-end	5	
Cash		5

Even the full historical journal would not distinguish these two contracts. Now impose a scenario in which the floating rate resets immediately to 7% for the whole next year. The fixed

loan produces interest of 5; the floating loan produces interest of 7. If the disclosure is also compatible with any fixed-rate fraction $\alpha \in [0, 1]$, future interest is

$$I(\alpha) = 100[0.05\alpha + 0.07(1 - \alpha)] = 7 - 2\alpha. \tag{1.1}$$

The historical accounts are compatible with the whole interval $[5, 7]$ of future interest. This is a range of feasible amounts. It becomes a probability distribution only when a distribution for α is supplied. Neither the number of generated interpretations nor a language model's confidence provides that distribution.

Journals remain essential because simulated economic events must have coherent accounting consequences. However, the simulated state must also contain contract terms, maturities, customer or borrower cohorts, collateral, unrecognised commitments, and management decisions. These determine what the journals will be under a changed economy. Reconstructing every historical transaction is unnecessary when a smaller state preserves the responses needed for the intended scenarios. Establishing that smaller state is an explicit modelling task, examined in Chapter 6.

The analogy with competing legal interpretations is useful at the level of representation. Several readings may survive the evidence and require separate reasoning. In accounting, some alternatives concern the meaning of a disclosure; others concern undisclosed facts or future behaviour. The system must identify which kind of uncertainty is involved. A known contractual term must not become an arbitrary interpretive choice merely because another choice is convenient for a simulation.

1.2 Users, questions and deliverables

An accounting analyst needs to inspect a number in its original table, verify its entity, currency and period, and see how it reconciles with the notes. A financial modeller needs to distinguish a reported balance from assumptions about its composition. A language engineer needs deterministic semantics and tests. A researcher needs to compare explanations under the same scenario, including explanations that fit history equally well.

The first release must answer the following complete workflow. An analyst loads a set of reports and specifies an information cutoff. AccDSL preserves the original documents and extracts source-linked observations. It identifies reconciliation failures and constructs an explicitly bounded family of compatible histories. The analyst declares additional assumptions where simulation requires them. The system executes each requested scenario and returns statements, journal traces, explanation-specific distributions or bounds, and unresolved limitations. When a later filing arrives, the analyst can compare the revised interpretation with the earlier information set.

Requirement	Required behaviour
R1 Evidence	Every observed amount or narrative assertion retains its document version, location, extraction method and original representation.
R2 History	Queries distinguish economic time, reporting period, publication time and database ingestion time; restatements remain recoverable.
R3 Meaning	Entity, perimeter, dimensions, currency, measurement basis, standard version and accounting concept accompany values.

Requirement	Required behaviour
R4 Ambiguity	Discrete alternatives and continuous unknowns retain their joint constraints across accounts and periods. Missing values never mean zero.
R5 Accounting	All committed journals balance in the book's functional currency, and balances reconcile to explicitly classified movements.
R6 Economics	Contracts, exposures and behavioural assumptions drive events. Accounting identities alone never determine scenario responses.
R7 Results	Conditional results survive by interpretation and scenario. Weighted results require declared, normalised probability measures.
R8 Explanation	A report cell can be traced to journals, events, rules, assumptions and evidence, including consolidation adjustments.
R9 Extensibility	Standard, industry and company packages extend shared types through versioned interfaces, with explicit unsupported cases.
R10 Reproduction	A saved run records source versions, model versions, parameters, calendars, seeds and numerical conventions.

These are acceptance requirements. Python, SQL storage, a particular parser, and a particular constraint solver are implementation choices. Initial support will cover a non-financial company with cash, receivables, payables, inventory, fixed assets, ordinary debt and ordinary equity. Banks and insurers must be representable in the core data model from the outset, but their executable measurement and behavioural packages follow dedicated milestones. Until a package supports an instrument, the result is explicitly incomplete.

1.3 What constitutes a useful first implementation

The first demonstrator should reconstruct three reporting dates and the intervening annual and interim flows for one company, including at least one restatement or revised comparative. It must preserve a borrowing-composition ambiguity and execute the fixed/floating example through both journals and all affected statement lines. A synthetic company supplies known hidden history; a public company supplies realistic disclosure and extraction problems. Their roles differ: the synthetic case tests preservation and rejection of known alternatives, while the public case tests traceability and practical coverage.

Success is a reproducible chain of evidence and computation, with identified gaps. A convincing-looking complete ledger reconstructed from incomplete disclosures would fail this product contract if its assumptions were hidden. Forecast accuracy requires a later evaluation with historical information cutoffs and suitable baselines; it is not established by the first demonstrator.

Chapter 2

A complete company and a real disclosure

2.1 What a balance must remember

Northstar Components is a synthetic seller of one homogeneous good. Its disclosures, transactions and contract terms are specified together, so we can check whether reconstruction preserves a known possible history. All amounts in this example are currency units, not thousands. The company has no foreign currency, subsidiaries, leases, returns or other comprehensive income. These are declared features of the example, not assumptions to impose on an imported company. Its source package and expected results are supplied in `fixtures/northstar.json`.

At the beginning of 2024, Northstar holds cash of 60, receivables of 20, inventory of 30 and plant costing 100 with accumulated depreciation of 20. It owes suppliers 20 and lenders 100. Share capital is 50 and retained earnings are 20. Thus assets of 190 equal liabilities of 120 plus equity of 70. Inventory consists of six units costing 5 each. The existing plant depreciates by 10 a year, with no residual value. The loan remains outstanding throughout 2024; both possible contract types charge 5% during that year.

The six months ending June and December provide two successive observation periods. Here the chosen accrual convention assigns one half-year to each. Cash-flow dates within a half-year are not recovered from a period aggregate; the synthetic history supplies a feasible ordering in which inventory is purchased before delivery and customer receipts precede the payments they fund. More detailed liquidity questions require more detailed dates.

Movement	First half	Second half
Delivered sales on credit	110	150
Customer collections	110	135
Inventory purchases on credit	70	85
Supplier payments	65	80
Cost of goods delivered	75	90
Payroll paid	20	22
Plant purchased at period end	10	12
Depreciation	5	6
Interest accrued and paid	2.5	2.5
Tax expense	1.5	5.9
Tax paid	3.5	3.9
Dividends paid	4	5

Each row has an economic meaning before it has a posting. Delivery of the good transfers control and creates an unconditional receivable; invoicing is assumed simultaneous in this example. Collection settles that receivable. Purchases increase inventory and trade payables; paying the supplier settles the payable. Delivery separately removes inventory at cost. Plant purchases are capital expenditure, while depreciation consumes its service potential. The addition of 10 enters service after June and has a five-year life, producing one unit of second-half depreciation. The addition of 12 enters service after December and has a twelve-year life.

Tax is deliberately simple: 20% of positive profit before tax, with all these expenses deductible and no deferred tax. The first-half payment exceeds its tax expense by 2, leaving prepaid tax; that prepayment is used in the second half. A genuine corporate tax package would need jurisdiction, tax basis, losses, timing differences and payment schedules. The example's tax rule governs its own numbers only.

2.2 Recovering movements without inventing invoices

Suppose the annual report discloses sales of 260, cost of sales of 165 and the opening and closing working-capital balances, while the collection and purchase totals are hidden. Under Northstar's explicit no-write-off, no-acquisition and no-currency-effect scope, the three bridges are

$$35 = 20 + 260 - C, \tag{2.1}$$

$$20 = 30 + P - 165, \tag{2.2}$$

$$30 = 20 + P - S. \tag{2.3}$$

Here C denotes collections, P purchases and S supplier payments, all in currency units over the same year. Substitution gives $C = 245$, $P = 155$ and $S = 145$. These are identifiable aggregate movements within this model. They do not identify which invoice was paid on which day. If acquisitions or write-offs are admitted, their movement variables enter the same bridges and these totals may cease to be unique.

The loan remains different. Write F and V for fixed and floating principal. The source imposes $F + V = 100$, $F, V \geq 0$ and $0.05F + 0.05V = 5$. The last equality adds no information about the split. Both endpoints and every intermediate allocation survive. Northstar therefore has identified working-capital totals and an unidentified financing exposure in the same joint historical family. A solver must not turn its convenient witness $F = 0$ into the company's recovered contract.

The model's historical slots are one delivery, collection, purchase, supplier settlement, payroll, plant addition, depreciation, interest, tax payment and dividend per half-year. Each slot represents a classified aggregate event. The template for collection contributes $+C$ to cash and $-C$ to receivables; its state effect reduces the unpaid receivable pool by C. The purchase template adds inventory quantity $P/5$, inventory cost P and a payable P. A cash-balance identity by itself cannot substitute for either template.

2.3 The complete historical statements

The corrected statements follow from those movements. Temporary income and expense accounts close to retained earnings at each half-year boundary; annual income is the sum of the two periods' classified movements, excluding closing entries.

Balance	Opening	June	December
Cash	60	65	74.6
Receivables	20	20	35
Inventory	30	25	20
Plant, net	80	85	91
Prepaid tax	0	2	0
Total assets	190	197	220.6
Trade payables	20	25	30
Debt	100	100	100
Share capital	50	50	50
Retained earnings	20	22	40.6
Liabilities and equity	190	197	220.6

Annual revenue of 260 less cost of sales 165, payroll 42, depreciation 11 and interest 5 gives profit before tax of 37. Tax expense is 7.4 and profit is 29.6. Other comprehensive income is zero under the declared scope, so total comprehensive income is also 29.6. Equity moves from 70 to $70 + 29.6 - 9 = 90.6$ after dividends.

The cash-flow statement supplies an independent reconciliation. Collections 245 less supplier payments 145, payroll 42, interest 5 and tax payments 7.4 produce operating cash of 45.6. Investing cash is -22 and financing cash is -9, giving a cash increase of 14.6. The indirect route gives the same operating cash: profit 29.6 plus depreciation 11, minus the receivables increase 15, plus the inventory decrease 10 and the payables increase 10. The prepaid-tax balance is zero at both annual endpoints. The two cash routes use different information and therefore provide a useful check on classification.

2.4 A correction changes what was knowable

The original interim filing reported revenue of 120 and receivables of 30. Ten units of revenue had been recorded before delivery. Its tax expense was 3.5, profit 14 and equity 80. The annual filing subsequently corrected the interim revenue and receivable by 10 and recognised the excess tax payment as prepaid tax of 2. Corrected interim profit is 6 and equity is 72, as above. The original statement remains an accurate record of what was published, even though its revenue recognition was wrong under the stated delivery rule.

For the temporal fixture, the original report becomes available on 1 August 2024 and is ingested on 3 August. The correction and annual report become available on 2 March 2025 and are ingested on 5 March. An investor-information query as of September 2024 returns 120. The same query as of 3 March 2025 returns 110 when evaluated using the subsequently acquired archive. A query that additionally asks what the database contained on 4 March still returns 120. Publication and ingestion answer different questions.

The annual total 260 and corrected interim total 110 share revision group `v2`, so second-half revenue is $260 - 110 = 150$. Subtracting the original 120 would yield 140, but that result mixes incompatible revisions. The query must reject the subtraction unless a reviewed compatibility relationship supports it. A later date alone does not establish such a relationship.

2.5 One economic scenario, several complete futures

For 2025, assume demand for 30 units at price 10, unit purchase cost 5 and a target closing inventory of eight units. Opening inventory is four units, so the purchase decision is $30+8-4=34$ units, costing 170. Deliveries cost 150, leaving inventory of 40. Collect 90% of current sales and 10 of old receivables: $270+10=280$, leaving receivables of 55. Pay old suppliers 30 and 130 of the new purchases, leaving payables of 40. Payroll is 50, capital expenditure of 20 occurs at year-end and dividends are 8.

These decisions are inputs to a conditional forecast. They are not estimates of demand or optimal management responses. No new borrowing is available. Under the existing asset schedules, depreciation is $10+10/5+12/12=13$. The new year-end asset begins depreciation in the next period. Let $\alpha = F/100$. With the floating rate resetting to 7%, interest is $I = 7-2\alpha$. Profit before tax is $300-150-50-13-I = 80+2\alpha$, giving tax $16+0.4\alpha$ and profit $64+1.6\alpha$.

2025 statement amount	All fixed	All floating
Revenue	300	300
Interest expense	5	7
Tax expense	16.4	16
Profit and comprehensive income	65.6	64
Operating cash flow	48.6	47
Investing cash flow	−20	−20
Financing cash flow	−8	−8
Closing cash	95.2	93.6
Receivables / inventory / net plant	55 / 40 / 98	55 / 40 / 98
Assets	288.2	286.6
Payables / debt	40 / 100	40 / 100
Equity	148.2	146.6

The joint symbolic result is cash $93.6+1.6\alpha$, profit $64+1.6\alpha$ and equity $146.6+1.6\alpha$, with the same $\alpha \in [0,1]$ throughout. Selecting the smallest cash and largest equity independently would describe no feasible company. A requested distribution requires a measure over that same historical family; the accounting equations do not supply one.

2.6 Apple: source labels do not identify exposure

The real reference package is Apple's 2024 Form 10-K and first-quarter 2025 Form 10-Q, for fiscal dates 28 September and 28 December 2024 respectively (Apple Inc., 2024, 2025). The annual balance sheet also supplies 30 September 2023. The annual filing is archived as the company's PDF from AnnualReports; the interim filing is archived from SEC EDGAR with its inline XBRL. Apple's Note 1 describes a 52/53-week year ending on the last Saturday of September. Calendar-year assumptions would therefore misstate the periods.

The package inventory preserves every annual PDF page and every interim top-level section, all thirteen annual notes and all ten interim notes. It records internal cross-references and externally incorporated material, including the proxy statement, as distinct obligations. Statement and debt facts below are structured; policies, other notes and exhibits retain their source locations and specific modelling obligations. This is complete inventory coverage of the acquired files, with partial semantic formalisation. It does not claim to reconstruct every Apple contract or to have acquired every incorporated external document.

Apple's annual balance sheet reports assets 364,980, liabilities 308,030 and equity 56,950, in millions of US dollars. Note 9 reports term-debt principal 97,341, unamortised discount/premium and issuance costs of −321, and hedge accounting adjustments of −358. Thus its carrying amount is $97{,}341 - 321 - 358 = 96{,}662$, split into current 10,912 and non-current 85,750. Fair value is separately disclosed as 88.4 billion. None of these three measurements may replace another merely because a table calls it debt.

Note 9 identifies the notes as fixed-rate, then states that interest-rate swaps effectively convert a portion to floating rates. Note 4 reports interest-rate contract notional of 14,575 at the annual date. A notional is neither a liability carrying amount nor sufficient information for an exact interest sensitivity: reset timing, remaining periods, curve conventions and hedge coverage still matter. The cross-reference connects the derivative to the borrowing without erasing either contract.

At the interim date, total term-debt carrying amount is $10{,}848 + 83{,}956 = 94{,}804$. Cash is 30,299, total assets 344,085, liabilities 277,327 and equity 66,758. The interim debt note's rounded "94.8 billion" and the statement's 94,804 million can coexist; their precision differs. Treating them as conflicting exact numbers would create an artificial infeasibility.

To demonstrate a source-to-forecast calculation, freeze the interim balance sheet and apply a *marginal* financing stress for a following 13-week period: a 200-basis-point rate increase, no principal change, no offsetting hedge beyond the explicitly selected exposure, immediate cash settlement and no tax effect. An assumed effective floating exposure $E = 14{,}575$ million produces $0.02 \times (13/52)E = 72.875$ million additional interest. This exposure uses the annual notional as a conditional input; it is not asserted to be the interim net sensitivity. Under that condition, debit interest expense and credit cash by 72.875. The resulting marginal statements have cash 30,226.125, assets 344,012.125, liabilities 277,327, equity 66,685.125, profit and comprehensive income of −72.875, and operating cash flow of −72.875 relative to the frozen baseline. The balance-sheet difference equals the equity difference exactly.

If exposure is instead only restricted to $0 \leq E \leq 14{,}575$, the conditional added interest lies in $[0, 72.875]$ before monetary rounding. That restriction is a stress assumption and is not claimed to bound every real Apple exposure. This real case verifies evidence, context, mappings and a coherent marginal propagation. Northstar supplies the complete-company forecast; a complete Apple forecast remains dependent on its unmodelled operations and undisclosed terms.

Chapter 3

Representing a company's historical record

Northstar's corrected receivable and Apple's several debt amounts show why the source record must retain more than a label and a number. The procedures below determine which assertions enter a model and which remain unresolved.

3.1 The record is a collection of assertions

The original record includes financial statements, notes, accounting policies, auditor opinions, management discussion, segment tables, security terms, regulatory returns and subsequent announcements. A complete ingestion process accounts for every page and every extracted block. A complete financial model is a stronger claim: it would require economically relevant information that public reports may never disclose. AccDSL therefore reports document coverage, semantic coverage and simulation coverage separately.

Each source span receives a disposition: structured observation, narrative assertion, relationship, policy evidence, contextual material, duplicate, or unresolved extraction. "Unresolved" remains visible. A commitment buried in a note is not discarded because it has no current balance-sheet line. An auditor's qualification is retained as evidence about the record; it does not become an automatic numerical haircut. Management's forecast is a dated assertion with an author and modality, not an observed future outcome.

An observation is an immutable record with the fields below. Values inferred from observations are separate records, linked by a derivation. This avoids overwriting a company's statement with an analyst's preferred reconstruction.

Field group	Minimum contents
Identity and source	Observation ID; document hash; filing identifier; PDF page and bounding box or HTML/XBRL locator; literal text; extraction version.
Reporting context	Legal entity; reported group perimeter; statement or note; instant or interval; standard and taxonomy versions; dimensions such as segment, product, geography, seniority or measurement category.

Field group	Minimum contents
Amount	Original string; parsed number; sign interpretation; unit; scale; currency; precision or rounding interval; missing-value category.
Meaning	Source label; taxonomy QName when present; one or more proposed concept mappings; whether the assertion is reported, estimated or conditional.
Timing	Economic validity when known; reporting period; publication and availability timestamps; ingestion timestamp; correction or supersession links.
Reliability and review	Extraction confidence, reviewer decision, source status and known contradictions, all distinct from a probability over histories.

A narrative assertion adds actor, predicate, object, scope, quantifier and modality. For example, "substantially all debt is fixed rate" has a source span and a qualitative quantifier. It does not assert $\alpha = 1$. A reviewed interpretation can introduce a numerical restriction such as $\alpha \geq a$, but the threshold a remains an analyst assumption unless the source defines it. Boilerplate policy language also does not prove that a company holds a particular instrument.

3.2 Different names, different measurements

An accounting concept is more than a row label. "Debt" can refer to legal principal, amortised carrying amount, undiscounted contractual cash flows or a fair-value disclosure. These amounts can all be correct at the same date. Likewise, "revenue" may be reported gross or net, a segment measure may differ from the consolidated measure, and an alternative performance measure may exclude selected expenses. A mapping must state its relationship: equivalent meaning, component of, broader than, presentation reclassification, or measurement transformation.

The common vocabulary will preserve source concepts as first-class identifiers. Company extensions may inherit a broad economic category without being declared identical to a standardised concept. For instance, an extension can be a component of borrowings while retaining an unresolved split between lease liabilities and other financing. Many-to-many mappings are allowed; each mapping has applicable dates, dimensions and evidence. An aggregation mapping does not silently become a measurement conversion.

Standards differ in recognition, measurement, presentation and disclosure. Consequently, conversion between IFRS, US GAAP and Chinese Accounting Standards for Business Enterprises (ASBE) is not generally a dictionary lookup. The original basis remains preserved, and any comparison basis is a separate derived book. The same principle applies to tax and prudential capital. A tax loss carryforward, accounting deferred tax asset and regulatory capital deduction can concern the same economic loss while belonging to different measurement systems.

3.3 Time, revisions and information availability

At least four times matter. A machine may enter service in November; its cost appears in the accounts for the year ending December; the annual report is published in March; and AccDSL ingests the report in April. A later report may revise the December comparative. These times answer different questions and must remain distinct.

The storage model uses validity intervals for economic assertions and transaction intervals for database versions, following the distinction in Snodgrass and Ahn (1986). Publication and availability timestamps are additional domain fields. Database transaction time records what AccDSL contained; it cannot substitute for what investors could have known. An "as known on" query applies an explicit availability rule and does not include a later restatement simply because its economic period is earlier.

AccDSL supports three named views: the original filing view; the latest restated view, with a stated information cutoff; and a harmonised research view, with explicit transformations. No view changes the stored source facts. Corrections have links to the replaced assertion and a reason. Where two sources conflict without a clear supersession relationship, both survive with a conflict record.

Instant values and interval flows have different types. A balance at 30 June is not a six-month flow. Calendar intervals use half-open boundaries $[t_0, t_1)$ internally; a filing period ending 30 June maps to the next boundary at 1 July in its reporting calendar. Quarter-only earnings may be derived from year-to-date values only if the periods, basis, perimeter, units and revision vintage are compatible. The derivation retains both source observations and their correlated rounding uncertainty. Fifty-two/fifty-three-week calendars, stub periods and reporting-date changes require explicit calendar objects.

Accounting-policy changes, estimate changes and error corrections also have different temporal effects. The applicable rule package decides whether a change is prospective, requires restatement, or adjusts opening balances. The engine never infers that effect from a generic "revision" flag.

3.4 The company boundary changes

Legal entity, reporting group and economic exposure are separate dimensions. Acquisitions, disposals, changes in control, joint arrangements, special-purpose vehicles and non-controlling interests can change the reporting perimeter. A group's increase in property need not be capital expenditure; it may reflect an acquisition, a currency translation or a revaluation. The bridge between opening and closing property must identify these movements before a scenario model can extrapolate investment.

The company graph therefore stores dated ownership, control and counterparty relationships. Control is an accounting conclusion with supporting evidence, not a rule that blindly consolidates every majority-owned entity. Group reporting combines entity books with a separate adjustment and elimination book. Intragroup receivables and payables require matching counterparties, currencies and dates. Unmatched eliminations remain unresolved; a balancing "plug" is not a recovered transaction.

Acquired opening balances, goodwill, purchase-price allocation, divested operations and non-controlling interests each have their own event types. Analyses can preserve reported scope or define a constant perimeter, but the latter is a derived modelling assumption, not the original record.

3.5 Further problems that the representation must handle

Several difficulties extend beyond differences in standards, industries and company labels. Recognition and settlement occur on different dates: a sale, invoice, cash receipt and impairment can belong to four periods. Net balances hide gross flows: repeated borrowing and

repayment can leave the same closing debt. Netting can conceal offsetting exposures and liquidity needs. Different contractual maturities and repricing dates can share the same current carrying amount. Guarantees, undrawn credit facilities, litigation and covenants can be economically important before an asset or liability is recognised.

Foreign currency introduces transaction currency, functional currency and presentation currency, together with different translation conventions for stocks and flows. Currency translation effects must not be mistaken for cash receipts. Inventory and depreciation depend on vintages or lots; an aggregate stock and average age may be insufficient under a shock. Tax bases can evolve differently from carrying amounts. Deferred tax recognition requires assumptions about recoverability, rather than multiplication by a tax rate alone.

Public information is selectively disclosed and rounded. Duplicate XBRL facts may agree, overlap within rounding bounds or contradict one another. A dash can mean zero, immaterial, unavailable or inapplicable. Tables can mix units, span pages and embed signs in parentheses. Text and table references can be wrong in the source itself. AccDSL retains these distinctions and gives an analyst a local repair path without pretending that extraction confidence settles economic interpretation.

Finally, strategic behaviour matters. A firm may refinance, cut investment, raise equity, change prices, breach a covenant or fail. A bank may preserve a capital ratio by shrinking assets; an insurer may change reinsurance or asset allocation. Such responses cannot be deduced from the balance-sheet identity. They require explicitly calibrated or assumed policies, and alternative policies must remain separate when the available record does not distinguish them.

3.6 Account for a disclosure before translating it

The importer first freezes a filing package: documents, byte hashes, reporting dates, availability evidence and incorporated references. It compares the document's contents page and statement index with detected section and note headings. A missing heading, table continuation or referenced exhibit creates an obligation to resolve; successful text extraction cannot discharge it. Each PDF page and each HTML content unit then has a stable locator and a disposition. Child units such as a table or note belong to a parent page/section; the importer records overlap instead of counting the same content twice.

Three denominators remain separate. Acquired-document coverage counts source units with a disposition, including unresolved units. Semantic coverage counts identified assertions whose meaning and context have been reviewed. Simulation coverage counts the operations and exposures required by a specified query that have supported state and rules. The first can be complete while the other two are partial. An assertion inventory can itself be incomplete; until the page/section reconciliation is reviewed, semantic coverage is a lower-bound inventory rather than a percentage of all financial meaning.

Cross-reference traversal is a work-list algorithm. Start with the filing's statement and note references, resolve each target against the acquired manifest, add newly discovered targets, and stop when no unvisited target remains or a declared resource limit is reached. Document hash plus target locator prevents cycles from expanding forever. A target has one of four results: resolved, duplicate of a resolved source, unavailable, or ambiguous. The last two remain obligations. Apple's proxy and prior-filed agreements show why closure over acquired files is weaker than closure over incorporated disclosures. A manifest may declare a bounded acquisition scope, but must show these outstanding references beside its completeness status.

A narrative translator records the actor, predicate, object, quantifier, modality, applicable entities/dates and qualifications before proposing any constraint. "All notes are fixed-rate"

constrains contractual coupon type; it does not constrain the combined note-and-swap exposure. "May repurchase" creates an authorised action with limits, not an obligatory payment. "Substantially all" remains qualitative unless a separately reviewed assumption gives it a threshold. A conditional guarantee becomes an obligation with a trigger and exposure, not a present liability merely because its maximum amount is known. Unsupported predicates retain their literal text and links to affected balances. The reviewer accepts the translation separately from accepting that the text was extracted correctly.

3.7 Mappings must establish a financial relationship

A mapping record identifies source concepts, target concept, relationship, applicable entity/basis/period/dimensions, evidence, reviewer and any required transformation. Equivalence permits substitution only when all these contexts match. A component relationship permits inclusion in a sum only through a declared disjoint partition. A broader concept supplies containment, not an equality. Reclassification changes presentation while preserving the stated population and total; measurement transformation must supply an actual formula and inputs. These relationships cannot compose indiscriminately: component followed by equivalence remains component, while two measurement transforms compose only if their intermediate bases and information dates agree.

For example, Northstar's product revenue might split into 180 domestic and 80 overseas, while its customer channel split is 200 wholesale and 60 retail. Both classify the same 260. Adding all four numbers gives 520 and is invalid. The partition records therefore share a population ID but have different axes. To remove an axis, `sumOver` requires evidence that members are disjoint and exhaustive for that context. Where the source supplies only named segments, an explicit unallocated member carries the residual; it is not assumed zero. Geography-by-product cells cannot be recovered from the two marginals without further constraints or assumptions.

Apple supplies another counterexample: its vendor non-trade receivables arise from component sales to manufacturing vendors, whose gains reduce product cost of sales when the final products are sold (annual Note 4, p. 38). They cannot be relabelled as ordinary revenue-generating trade receivables without changing meaning. Likewise gross derivative assets and liabilities cannot be silently netted. Company extensions can define these distinctions and dated presentation changes; they cannot override mandatory recognition rules in a selected standards package.

3.8 Selecting original and revised assertions

The query procedure first filters assertions by the declared availability cutoff, then by an optional database-time cutoff. A day-only filing timestamp is conservatively available at the next local midnight in its named timezone; any more precise rule needs evidence. Within the eligible set, explicit supersession removes only the targeted assertions in matching contexts. The superseding assertion must itself be eligible. Follow the acyclic graph transitively; retain all maximal assertions if two eligible corrections conflict. File order and ingestion recency never decide which is true.

The original view selects the initial assertion chain in a named filing; the latest-available view selects eligible maximal versions; the harmonised view adds reviewed transformations to those selections. A revision to revenue does not automatically revise every note or change the company's accounting basis. An estimate change takes effect in future measurements from its specified date. An error correction may change a comparative and an opening balance bridge, each as a new assertion with its own lineage. Northstar's explicit query fixture in Chapter 2 checks all these clocks without assuming that the most recently stored row is always the appropriate history.

Chapter 4

What the literature contributes

The research problem crosses accounting, databases, programming languages, econometrics and document understanding. Each field resolves part of it. The review below distinguishes results that can be adopted directly, engineering ideas that need adaptation, and evidence that does not establish the intended product capability. Source versions and technical anchors are recorded in the accompanying research ledgers. The search includes work outside the original collection, with public-index searches and backward and forward citation searches through 2 October 2026. It is a scoped design survey, rather than a claim to enumerate every paper in these fields.

4.1 Accounting structure and company dynamics

Nehmer and Robinson (1997), especially Sections 2–5, supplies a useful algebraic starting point: balances occupy a structured state space, transactions change that state, and aggregation relates detailed accounts to reports. This supports a distinction between conservation constraints and permitted transactions. An entry can conserve the accounting total and still be an inadmissible transaction. AccDSL adopts that distinction in its posting validator. Its particular event vocabulary, evidence model and inverse-history semantics are new design decisions, not results established by the algebraic paper.

Doshi (2024) develops an accounting DSL with time, transaction logs and correction mechanisms. The relevant material is in Chapters 4 and 7–9 of the 2024 thesis; the original collection's filename misleadingly suggests 2026. It demonstrates why corrections and dates belong in the language rather than in informal comments. The thesis does not establish recovery of an entire company history from public reports or inference over economically distinct histories. Babcinețchi et al. (2022), pp. 342–345, offers a much smaller accounting language and grammar proposal. It is useful as a syntax precedent, but it provides insufficient semantics or evaluation to govern AccDSL by itself.

Shahnazarian (2004) connects accounting dynamics with behavioural and econometric equations. Sections 2.3–2.8 describe the accounting relationships; the optimisation and econometric discussion in Chapter 3 and the data and evaluation material in Chapters 4 and 8 show how the model is used with standardised Swedish corporate data. This is an important precedent for simulating companies under changed conditions. Its standardised tax-data starting point differs from heterogeneous public-report extraction, and its working-capital funds formulation should not be imported as a general modern cash-flow statement implementation. The transferable idea is the explicit separation of accounting evolution and behavioural assumptions.

Pierson (2020), Sections 3–4 and the reporting structures, makes that separation operational in system dynamics. Receivables, payables and wages accumulate differences between accrual and payment flows; reporting-period accumulators turn instantaneous flows into statements. AccDSL can use these structures as examples of coherent movement schedules. The model's payment delays, inventory approximations and simplified equity treatment remain modelling choices. They are not universal defaults for companies or evidence that an aggregate state is sufficient for all shocks.

Bank models demonstrate why journal generation needs economic state. Halaj (2016), Section 3, combines funding shocks, fire sales, loans, securities and risk constraints. The homogeneous portfolios and specified funding process make the model tractable, but limit instrument-level interpretation. Behn et al. (2019), Section 2, makes managers' choices over asset growth, debt, risk and payouts depend on capital and liquidity constraints. These models motivate interfaces for management policy and regulatory calculations. Their parameterisations must be calibrated and evaluated for the target application; passing their accounting identities would not validate an AccDSL forecast.

Two important neighbouring traditions remain explicit coverage qualifications. McCarthy's 1982 REA model is a foundational candidate for the relation between resources, events and agents; Caiani and co-authors' stock-flow-consistent agent-based benchmark is a candidate for simulation across interacting sectors. Verified primary PDFs could not be obtained in this retrieval campaign after institutional and publisher attempts. Their records are retained in the omission register, and no technical result from either is used to justify the design. AccDSL's event and counterparty records are specified and justified directly below. A later multi-company equilibrium engine would require expansion of the stock-flow-consistent literature before implementation.

4.2 Incomplete information and provenance

Financial disclosures are incomplete databases in a substantive sense. Multiple underlying records can yield the same published observations. Libkin (2014), Sections 2–5, explains the semantics of incomplete databases, labelled nulls, conditional tables and certain answers. Repeated occurrences of the same unknown must share a value. Conditional tables can represent dependence between alternatives that independent missing cells cannot express. This is directly relevant to a debt note and a cash-flow table that refer to the same unknown repayment.

AccDSL adopts shared symbolic unknowns and conditions. It extends the database idea with accounting constraints, continuous amounts, temporal transitions and economic event models. A "certain" answer means a statement that holds for every history in the declared feasible family. The family itself may omit unknown mechanisms; certainty is therefore always relative to a model scope. Imieliński and Lipski's original 1984 paper is a foundational backward lead whose publisher copy was inaccessible in this campaign. The inspected Libkin paper supplies the technical discussion actually used here.

Green et al. (2007), Definitions 3.1–3.2 and Section 4, treats provenance as annotations that compose through relational operations. Combining sources and alternative derivations can therefore have formal semantics, rather than being a list of unrelated citations. This is useful for report aggregation and explanation queries. Positive relational operations covered by provenance semirings do not by themselves specify every operation AccDSL needs: negation, nonlinear valuation, optimisation and probabilistic conditioning require their own derivation records. Accordingly, the implementation starts with a typed dependency graph and may use semiring annotations for supported relational subqueries. A dependency edge is not a posterior

probability.

Snodgrass and Ahn (1986) distinguishes historical validity from the time facts are stored. The archived paper is *Temporal Databases*, by Snodgrass and Ahn, not the later Jensen–Snodgrass paper initially returned by a search. Its snapshot, rollback, historical and temporal examples support the need to recover both old business facts and old database beliefs. AccDSL adds public availability time because a database insertion date cannot establish whether a historical investor could have seen a filing.

4.3 Inverse problems, identification and probability

Stuart (2010), Section 2.2, starts from an observation equation $y = G(u)$ and distinguishes inversion, regularisation and Bayesian inference. The discussion makes explicit that a regularising norm or prior is an additional assumption. AccDSL has a similar observation problem, although its unknowns include discrete accounting interpretations and event structures as well as continuous parameters. The infinite-dimensional well-posedness theorems in that paper are not automatically applicable to this mixed, potentially discontinuous model.

The relevant issue is often identification rather than computation alone. Giacomini and Kitagawa (2021), Sections 2.1–2.6 and Theorem 1 of the archived 2020 working version, shows how a class of priors can preserve uncertainty within an identified set. A likelihood that is flat over the unresolved part does not learn the conditional prior over that part. Posterior lower and upper probabilities summarise a class of posterior beliefs. The paper's regularity conditions and application to structural vector autoregressions do not prove that an arbitrary accounting program admits the same algorithms or guarantees. The design borrows the distinction between learnable quantities and assumption-dependent distributions; the simple accounting bounds in Chapter 6 are derived directly.

Hoeting et al. (1999), equations (1)–(3), gives Bayesian model averaging as a weighted sum of model-conditional distributions, with weights determined by model priors and marginal likelihoods. Sections 3 and 5 discuss computation and prior specification. This supports an optional weighted mode, provided AccDSL specifies the likelihood, priors and candidate model space. It does not justify giving equal mass to every syntactically different journal reconstruction. Two encodings of the same economic history must not gain extra weight by duplication. Model averaging also remains conditional on the considered model family, rather than repairing missing mechanisms.

Scenario semantics need similar care. Pearl (2009), Section 3.2.1 and equations (5)–(6), distinguishes an intervention that replaces a structural equation from ordinary conditioning. AccDSL will distinguish a supplied exogenous path, an observational conditional forecast and an intervention. A rise in rates imposed by a stress test is not automatically distributed as the collection of historical periods in which rates happened to be high. Identifying a causal effect requires assumptions beyond reproducing accounts.

4.4 Languages for rules and contracts

van Deursen et al. (2000) surveys DSL benefits, risks and implementation options. Mernik et al. (2005), Sections 1.1 and 2.2–2.5 of the archived CWI report, treats application libraries as serious competitors and distinguishes deciding, analysing, designing and implementing a DSL. Its implementation decision tree explicitly includes embedding in a host language. The literature therefore supports a conditional design decision: a DSL is attractive when domain

constraints and repeated notation justify it, but an ordinary typed library may be cheaper and equally expressive. AccDSL will implement one shared representation accessible from both a Python API and a textual language, so syntax does not become a prerequisite for testing the accounting model.

Merigoux et al. (2021), Sections 3–4 and Figures 8–9, provides Catala's scopes, exceptions, default calculus and translation into a conventional functional target. Its explicit empty and conflict outcomes are valuable precedents for accounting rules with exceptions. They are different from probabilistic uncertainty. AccDSL will preserve this distinction: conflicting applicable rules produce a diagnostic, while an evidence-supported alternative reading produces a conditional model branch. The paper's compiler correctness results do not establish correctness of our accounting rules or of an automatic translation from disclosures.

The inspected LegalMath code offers reusable ideas in its shared typed interpretation, partial observations, source references and validity/knowledge times. Its ordinary frontend execution selects a reading. AccDSL needs an additional execution mode that retains a family of readings, continuous unknowns and economic state. A bounded candidate generator is useful for search, but its candidate limit does not prove completeness.

Peyton Jones et al. (2000), Sections 3–5, builds complex financial contracts from combinators and separates their compositional meaning from a chosen valuation model. The ACTUS technical specification similarly separates contract states, scheduled events, state transitions and payoffs, with risk factors supplied through an interface (ACTUS Financial Research Foundation, 2026). These are strong reasons to represent contractual obligations independently of accounting classifications. Neither contract cash flows nor a derivative price determine the recognition, measurement and presentation rules of a complete IFRS or US GAAP book.

4.5 Extraction and financial reasoning

The document literature supports a staged ingestion process. PubTables-1M (Smock et al., 2022), in its canonicalisation procedure and evaluation section, distinguishes table detection, structure recognition and functional analysis. Its GriTS evaluation compares reconstructed table structure. TableBank (Li et al., 2020), Sections 3–4, uses document structure to obtain weak supervision for table detection and recognition. GTE and FinTabNet (Zheng et al., 2021), Sections 3–5, provide a financial-document setting and company-level dataset splits. These sources help select extraction components and evaluation units. Their table scores do not establish that a debt note has been interpreted correctly across years or that every off-balance-sheet exposure has been found.

The Docling report (Auer et al., 2024), Section 3, describes PDF parsing, page layout analysis and table structure recognition within a document-conversion pipeline. The locally archived version lists Christoph Auer first; this version difference is recorded in the catalogue. Docling is an ingestion candidate whose output must retain source locations and undergo accounting validation. Its code and model licences must be considered separately when packaging a deployment.

FinQA (Chen et al., 2021), Sections 3–6, provides questions with executable reasoning programs and supporting evidence. TAT-QA (Zhu et al., 2021), Sections 2–4, combines table cells and text spans with aggregation operations, and explicitly treats numerical signs in evaluation. DocFinQA (Reddy et al., 2024) extends financial questions to long documents and evaluates retrieval/context choices. FinBen (Xie et al., 2024) broadens the collection of financial tasks and model evaluations. These benchmarks can test retrieval and local numerical reasoning. Their question-answer format does not require a complete multi-period company representation, joint

alternative histories or scenario-dependent journal generation.

A directly relevant recent competitor is the auditable RAG architecture of Cosentino et al. (2026). Sections 3.1–3.2 describe persistent extraction, enrichment, retrieval and provenance records; Section 4.6 checks citation resolvability and failure recovery. Section 5 limits the interpretation of those results to the evaluated document-intelligence tasks. AccDSL should reuse the idea of persistent evidence and recoverable processing, while adding executable accounting and economic semantics. No performance comparison with that architecture has been run here.

4.6 Solvers and probabilistic execution

de Moura and Bjørner (2008), Sections 2–3, describes Z3's interfaces and theory-solving architecture. It supports choosing an SMT solver for logical choices and arithmetic constraints, but does not make arbitrary nonlinear simulation tractable. The first solver fragment is deliberately bounded: finite choices, exact linear arithmetic and selected calendar relations. Unsupported formulas remain in the representation with a solver diagnostic.

van de Meent et al. (2018) develops probabilistic programs through sampling, observations and inference; the inspected material includes the first-order language and weighted sampling discussion. Stan (Carpenter et al., 2017), its model language and sampler-output discussion, and Pyro (Bingham et al., 2019), Sections 2–3, show different ways to separate a probabilistic model from inference machinery. They are candidates for optional inference over an already-defined economic model. Exact accounting identities and discrete rule selection should not be turned into nearly satisfied likelihood penalties merely to fit an inference library. Nor should successful sampler execution be described as evidence that all interpretations were explored.

4.7 The remaining design problem

The combined literature provides algebra for journals, semantics for missing information, temporal storage, provenance, contract languages, rule languages, inference methods and extraction tools. The unresolved integration problem is to retain their distinctions while connecting them: a disclosure constrains economic histories; a history supplies contract and behavioural state; a scenario drives events; accounting rules produce journals; reporting rules render statements; and every step remains explainable.

The design is therefore an integration proposal with several local mathematical constructions. Its originality and empirical advantage have not been established by this survey. Its practical merit will depend on the acceptance cases, accounting review and comparative implementation work in Chapter 11.

Chapter 5

Architecture and reuse

5.1 One representation, several authoring methods

The architectural centre is a typed intermediate representation (IR), not a text parser. A document extractor, a human editor, a Python library and the AccDSL compiler all create the same records. This allows a modeller to begin with a reviewed structured file and later add convenient notation. It also allows source extraction to improve without changing accounting semantics.

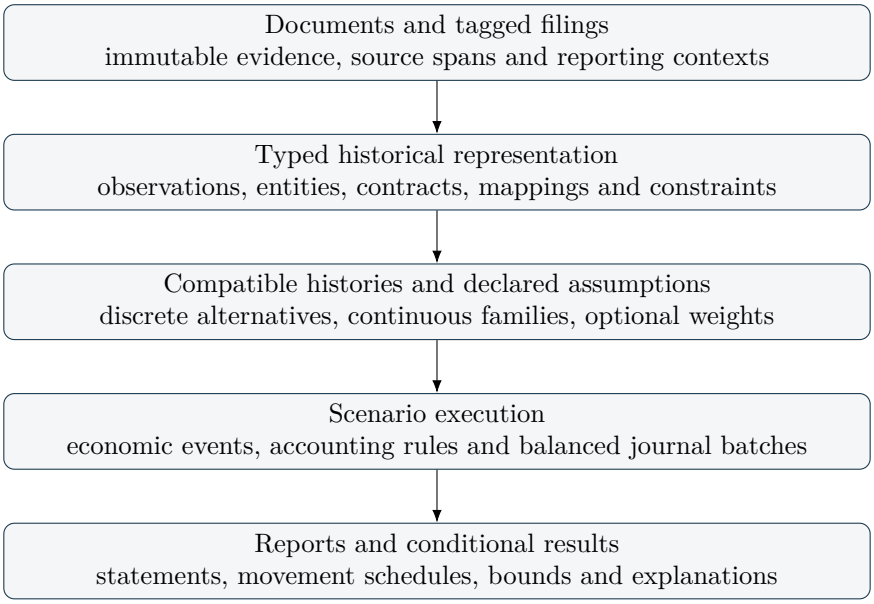


Figure 5.1: The simulation path. Every transformation records its inputs and assumptions; the source record remains available throughout.

Evidence storage and economic modelling must remain separate. If a company reports 100 of borrowing, a model can assume two loans of 60 and 40 without creating a new reported fact that the company disclosed those loans. The contract library consumes the assumed terms, the accounting module consumes the resulting events, and the explanation graph records the assumption at each affected result.

5.2 Why a DSL, and when another approach is better

The strongest baseline is a typed schema, SQL storage and ordinary Python code. That combination can implement the whole product. Its disadvantage is that repeated accounting requirements can become conventions scattered across functions: implicit currency conversion, missing period checks, hidden default values and duplicated rule selection. A bounded language makes these requirements part of typing and compilation. Its costs are language maintenance, diagnostics, migration and training.

Approach	Useful role	Limitation for this product
Typed schema and Python	Reference executor, flexible economic models, initial authoring API	Domain restrictions require disciplined APIs or a validator; arbitrary code complicates static analysis and provenance.
XBRL and OIM	Tagged evidence, contexts, dimensions, taxonomies and exchange	Reported facts and reporting relationships do not recover hidden economic histories or specify scenario behaviour.
Ledger languages	Recording known transactions, balance checks, export and differential testing	Alternatives, source interpretation and contract exposure need additional semantics.
Knowledge graph / SQL	Relationships, lineage, temporal queries and evidence navigation	A graph or schema alone does not define event execution or the probability meaning of alternatives.
Rules / Catala	Dated policy logic, defaults, exceptions, traceable rule evaluation	Ledger state, event scheduling and ensembles require an enclosing runtime.
Probabilistic program	Priors, likelihoods and inference over model parameters	Needs declared probability measures; cannot turn unspecified ambiguity into probability or supply accounting rules.
Spreadsheets	Familiar review and small transparent financial models	Versioned evidence, shared unknowns and branching histories are difficult to govern as the model grows.
AccDSL over shared IR	Common domain checks, reusable templates, explicit alternatives and deterministic posting semantics	Requires implementation and evaluation; broad notation without stable semantics would add cost.

The decision is to implement the IR, validators and Python reference API first, then the textual frontend. The same conformance cases must pass through both frontends. If the DSL fails to reduce ambiguity in representative authoring tasks, the IR and library remain useful. This is a reversible product decision, not a claim that a new language is inherently better than a library.

5.3 Open-source components

The following survey uses official repository descriptions, licence files and selected source interfaces archived on 2 October 2026. It is a source inspection, not an executed integration or performance evaluation. The machine-readable tool ledger records retrieval paths and available version metadata. Dependencies must be pinned when implementation begins; a branch name observed during this survey is not a release requirement.

Component	Proposed use and checked boundary	Licence / decision
Arelle	Parse and validate XBRL/iXBRL. Inspected `ModelFact`, context and unit accessors. Preserve taxonomy extensions and full contexts when importing.	Apache-2.0 core; first-choice adapter.
Docling	PDF layout, text and table conversion. Inspected converter interface; retain page coordinates and extraction settings.	MIT code; assess model licences separately. Candidate.
Table Transformer	Table extraction baseline associated with PubTables-1M; check finance-domain performance before selection.	MIT; repository marked archived in inspected metadata. Baseline, not a maintenance default.
Beancount	Posting and transaction data types; export selected concrete branches and compare balances.	GPL-2.0 project metadata; integration and redistribution choice needed. Optional adapter.
Ledger	Known-journal reference and export. Inspected transaction balancing code and manual date/assertion behaviour.	BSD-style licence; optional reference tool.
hledger	Alternative ledger import/export and accounting checks. Repository documentation inspected.	GPL-3.0 project metadata; optional process adapter.
Catala	Potential backend for deterministic policy modules; defaults and exceptions have useful semantics.	Apache-2.0; prototype only after core execution semantics stabilise.
ACTUS	Contract schedules, payoffs and transitions; external risk-factor interface. Verify event schedules against ACTUS tests.	Service Apache-2.0; technical specification CC-BY-SA-4.0. Candidate contract adapter.
QuantLib	Calendars, cash flows and valuation. Floating coupon source separates fixing dates and pricing. Accounting treatment remains external.	Modified BSD-style licence; candidate valuation adapter.
OpenFisca	Versioned tax-benefit formulas as a rule-system reference; inspected variable/formula metadata. Corporate-tax coverage must be supplied.	AGPL-3.0; optional, not the core runtime.

Component	Proposed use and checked boundary	Licence / decision
Z3	Finite alternatives, logical compatibility, linear constraints and conflict explanations.	MIT licence text; first solver candidate.
DuckDB	Local analytical queries over extracted facts and results.	MIT; candidate alongside SQLite transactional storage.
Stan	Continuous probabilistic models and diagnostics with an explicit target. Inspected model interface and paper.	BSD-3-Clause; optional inference backend.
Pyro / PyMC	Alternative probabilistic backends for model-specific inference. Pyro design paper and project descriptions inspected.	Apache-2.0 / Apache-2.0 licence texts; optional, no default selection.

The licence entries identify the inspected projects, not a legal conclusion about every combination of code, models and datasets. The default product architecture can communicate with optional tools through explicit adapters. This does not, by itself, settle redistribution obligations. That question is resolved when a concrete dependency and distribution method are selected.

XBRL's Open Information Model supplies standardised facts with concepts, entities, periods, units and dimensions (XBRL International, 2021). Those facts are an excellent ingestion source. AccDSL retains their original contexts and decimals rather than flattening them immediately into a company-by-year table. Calculation relationships can help validate reporting totals, but are not a general description of the transactions that generated those totals.

5.4 Responsibilities that AccDSL must implement

AccDSL owns the evidence and temporal model, typed accounting concepts, shared unknowns and alternatives, rule-package selection, journal validation, consolidation provenance and report explanations. It also owns the distinction between feasible sets and distributions. These responsibilities cannot be delegated implicitly to whichever parser or solver is convenient.

External components receive narrow requests and return typed responses. A contract adapter returns scheduled events, cash-flow quantities, state changes and source/model references; it does not directly mutate the ledger. A valuation adapter returns a measure, currency, date, method and numerical status; the accounting package decides whether and where to recognise the change. A solver returns feasibility or bounds with scope and completeness status. A language model proposes source-linked mappings or interpretations; it cannot commit an unsupported assertion as fact.

The architecture consequently permits deterministic reference execution, alternative solvers and later acceleration without changing the meaning of a historical observation or a journal entry. Performance work begins only after the same semantic cases pass through these interfaces.

Chapter 6

Historical compatibility and future simulation

6.1 Define what is observed before inverting it

Let D_k be the disclosures available under information cutoff k. Each numeric observation d_i has a context: entity or group perimeter, concept, measurement basis, dimensions, currency, period and precision. These contexts are part of the observation operator, not additional columns that can be discarded after extraction.

A candidate history h contains an opening economic and accounting state, events through the last historical date T, contract terms, accounting policies, consolidation relationships and latent quantities. Let M denote the declared model scope: its event templates, admissible rule versions and allowed unknowns. It excludes arbitrary events that the chosen model has no way to represent. The reporting operator $O_i(h)$ produces the quantity to which observation i refers, including its basis and perimeter.

Reported numbers are often rounded. If a number v_i is rounded to the nearest unit q_i, an illustrative admissible interval is $[v_i - q_i/2, v_i + q_i/2)$ under the corresponding tie convention. The actual rounding convention, scale and XBRL accuracy metadata govern the imported constraint. A disclosure with unspecified precision does not acquire a convenient tolerance silently. Write $\mathcal{I}_i$ for the justified observation interval, and $C_M(h)$ for the joint accounting, contractual and semantic constraints. Then the historical compatibility set is

$$\mathcal{H}_M(D_k) = \{h \in \mathcal{H}_M : C_M(h) \ \wedge \ O_i(h) \in \mathcal{I}_i \text{ for every applicable } i\}. \tag{6.1}$$

Qualitative assertions enter as reviewed logical conditions, rather than numeric intervals. Incompatible source contexts are never forced into the same equation.

Equation (6.1) distinguishes three outcomes. An empty set means that the observations and declared model are inconsistent. A singleton means that this model identifies a history at its chosen resolution. A larger set preserves unresolved alternatives. An empty set does not prove that the company's accounts are false: a wrong extraction, missing event type or incorrect perimeter can cause the inconsistency. A singleton does not prove that the model contains the real history.

6.2 Why published statements cannot identify all transactions

The simplest derivation uses fixed account mappings and a fixed basis. Let $z \in \mathbb{R}^n$ contain unknown historical movement amounts. For m disclosed aggregates, let $A \in \mathbb{R}^{m \times n}$ be the aggregation matrix and $y = Az \in \mathbb{R}^m$. If there is a nonzero vector v with $Av = 0$, then

$$A(z + v) = Az + Av = y.$$

Both movement vectors give the same reports whenever both satisfy the remaining constraints. Thus the feasible family is the affine set $z + \ker A$ intersected with those constraints. Accounting requirements can shrink this family, but do not generally eliminate it. For example, cash collections in a receivables bridge are constrained by

$$AR_T = AR_0 + \text{credit sales} - \text{collections} - \text{write-offs} + \text{other classified movements}.$$

If the note does not distinguish acquisitions, currency effects and write-offs, several collections paths may fit the same closing balance.

There is a second ambiguity beyond aggregation. In Equation (1.1), identical historical movements arise from different contractual reset terms. Even recovering z exactly would leave these terms unidentified. AccDSL therefore infers economically relevant states and contract terms alongside journal movements. A linear inverse solver over statement lines alone cannot solve this second problem.

6.3 The accounting state and its invariants

For one entity book with n_a accounts in one functional currency, let $b_t \in \mathbb{Q}^{n_a}$ be signed account balances, with debit amounts positive and credit amounts negative. Exact decimal amounts can be represented as rationals with a power of ten denominator. A journal batch j_t contains the signed functional-currency amounts posted by one atomic event or explicitly grouped event batch. Its balance condition is

$$\mathbf{1}^\mathsf{T} j_t = 0, \qquad b_t = b_{t^-} + j_t. \tag{6.2}$$

If $\mathbf{1}^\mathsf{T} b_0 = 0$, substitution gives $\mathbf{1}^\mathsf{T} b_t = \mathbf{1}^\mathsf{T} b_{t^-} + \mathbf{1}^\mathsf{T} j_t = 0$. Induction establishes conservation through every committed batch. Opening balances must therefore be supplied as a balanced, explained entry or a balanced imported state with unresolved composition explicitly identified.

This proof establishes a bookkeeping invariant. It does not establish that an amount belongs in the chosen account, that a contract exists, or that an accounting standard permits a treatment. Posting validation additionally checks account roles, ownership, currency conversions, applicability of rules and event preconditions.

The balance vector includes temporary revenue and expense accounts during the period. Period closing transfers their net balance to equity through balanced entries. A reporting function can include current profit in equity before closing, or use the closed book afterwards, but must not include it twice. Income and cash-flow reports use classified movements and exclude mechanical closing transfers from operating flows. Consequently statements are functions of balances, journals, contract state and reporting rules:

$$Y_{[u,v)} = R(b_{v^-}, J_{[u,v)}, X_{v^-}, \rho, [u, v)). \tag{6.3}$$

Here ρ specifies the reporting basis and perimeter, and X contains economic state beyond ledger balances. Cash-flow classification is part of R, not recoverable from the closing cash balance alone.

Each book has its own functional currency. Quantities such as shares, tonnes and foreign-currency units are typed separately and never added to this balance equation. A foreign-currency posting stores its original units and the functional-currency valuation with its rate source. Translation into a group's presentation currency and consolidation adjustments occur in a separate book under explicit rules.

6.4 Simulate events from economic state

For a chosen history h, the state at T is $X_T(h)$. It contains balances, open contracts, lots or cohorts, unpaid accruals, commitments and other quantities needed for future evolution. Let s identify a scenario definition, U_t^s its supplied economic drivers, a_t management's actions, and ε_t future innovations. A behavioural policy π chooses a_t using the permitted information $\mathcal{F}_t$:

$$a_t = \pi(X_t, U_{\le t}^s, \mathcal{F}_t; \theta), \tag{6.4}$$

$$e_{t+1} = \mathcal{G}(X_t, a_t, U_{t:t+1}^s, \varepsilon_{t+1}; \theta), \tag{6.5}$$

$$(j_{t+1}, m_{t+1}) = \mathcal{A}(e_{t+1}, X_t, r_{t+1}), \tag{6.6}$$

$$X_{t+1} = \mathcal{T}(X_t, e_{t+1}, j_{t+1}, m_{t+1}). \tag{6.7}$$

The generator $\mathcal{G}$ emits events, such as a sale, rate reset, payment or default. The accounting function $\mathcal{A}$ applies rule package r and returns journal postings plus measurement annotations m. The transition $\mathcal{T}$ updates economic and ledger state atomically. A default can therefore change the loan's legal status, expected recoveries, recognised allowance and cash collections through related, but distinct, records.

This ordering is a reference abstraction. Within a date, event dependencies and phases specify the actual order. Expectations used to measure an allowance or provision must be calculated from the information permitted at that date. They cannot use the realised future path of the simulation. An inner expectation model may use quadrature, analytic formulas or nested simulation; its approximation and seed belong in the measurement annotation.

Some scenarios specify a deterministic macroeconomic path while leaving idiosyncratic defaults random. Others supply a joint stochastic driver model. A risk-neutral measure used to value a derivative is distinct from the real-world measure used to simulate business outcomes. Every stochastic adapter declares its measure and conditioning information. The accounting function receives the resulting valuation with that declaration.

6.5 Preserve sets and distributions separately

Let $K_{h,s}$ be the probability law of future reports produced by the forward model conditional on history h, scenario s and its declared behavioural assumptions. A deterministic simulation is the special case of a point mass. When no probabilities over historical explanations are justified, the natural answer is a family of laws,

$$\mathcal{K}_{M,s}(D_k) = \{K_{h,s} : h \in \mathcal{H}_M(D_k)\}. \tag{6.8}$$

The system can report each explanation, bounds on a deterministic outcome, or lower and upper event probabilities across this family. Those are different result types. A range of conditional means is not a confidence interval, and an envelope of conditional quantiles is not itself a single predictive distribution.

If a defensible probability measure $\mu(dh \mid D_k, M)$ supported on $\mathcal{H}_M(D_k)$ is supplied, histories and report outcomes must have specified measurable spaces. Assume that $K_{h,s}$ is a probability kernel: it is a probability measure on report outcomes for each h, and $h \mapsto K_{h,s}(B)$ is measurable for each measurable event B. The predictive law is then

$$\mathbb{P}(Y^s \in B \mid D_k, M) = \int K_{h,s}(B)\, \mu(dh \mid D_k, M). \tag{6.9}$$

For discrete explanations this is the usual weighted sum. For continuous unknowns, integration uses the declared measure over those unknowns, not a count of solver solutions. Bayesian weights require a prior and a likelihood; expert weights remain labelled as elicited. A likelihood must not treat a statement total and all its components as independent repeated evidence. Deterministic equality constraints require a reduced parameterisation or proper conditional-measure treatment, rather than an undocumented Gaussian penalty with near-zero variance.

Suppose the fixed-loan explanation receives weight w and the floating-loan explanation receives $1 - w$, solely as an illustration. Under the rate shock,

$$\mathbb{P}(I = 5) = w, \qquad \mathbb{P}(I = 7) = 1 - w, \qquad \mathbb{E}[I] = 7 - 2w.$$

The expected expense is useful but hides the two different outcomes. For a general scalar output with finite second moment, put $m(h) = \mathbb{E}[Y \mid h]$. Write $Y - \mathbb{E}[Y] = (Y - m(h)) + (m(h) - \mathbb{E}[Y])$. The cross term vanishes because $\mathbb{E}[Y - m(h) \mid h] = 0$. Taking expectations of the squared expression gives

$$\mathrm{Var}(Y) = \mathbb{E}_h[\mathrm{Var}(Y \mid h)] + \mathrm{Var}_h(\mathbb{E}[Y \mid h]). \tag{6.10}$$

The first term is uncertainty within a chosen explanation; the second captures differences between explanations. This decomposition requires the measure μ. Without it, the system reports the family in Equation (6.8), rather than manufacturing a variance.

Weights themselves can be uncertain. For an admissible class $\mathcal{W}$ of measures, AccDSL can compute

$$\underline{p}(B) = \inf_{\mu \in \mathcal{W}} \int K_{h,s}(B)\, d\mu(h), \qquad \overline{p}(B) = \sup_{\mu \in \mathcal{W}} \int K_{h,s}(B)\, d\mu(h).$$

If every measure supported on the feasible histories is allowed and point masses are included, these bounds reduce to the infimum and supremum of $K_{h,s}(B)$ over histories: averages cannot lie outside the range of their integrands, and point masses attain, or approximate, the extremes. Narrower weight restrictions require an explicit justification and a separate optimisation problem.

6.6 Resolution and computational limits

For a requested horizon H and scenario family $\mathcal{S}$, histories may be grouped only when they have the same conditional law for all requested outputs:

$$h_1 \sim_{H,\mathcal{S}} h_2 \quad \Longleftrightarrow \quad K_{h_1,s} = K_{h_2,s} \text{ on those outputs for every } s \in \mathcal{S}.$$

This definition explains why individual sales can sometimes be replaced by cohorts, while fixed and floating loans usually cannot be merged for an interest-rate stress. Exact equivalence can be difficult to prove. A grouping justified only by approximate numerical agreement carries an error criterion and is not labelled exact. Source provenance remains available even when execution shares a common state.

Bounds require comparable honesty. A linear program can provide certified extrema for a supported linear model. A set of feasible sampled histories provides observed extrema: its

minimum is no smaller than the true infimum, and its maximum is no larger than the true supremum. It therefore supplies an inner exploration of possibilities, not a certified outer bound. Interval methods may provide outer bounds that no single history attains. Every result records which of these meanings applies.

The initial implementation offers exact bounded choices and linear constraints, symbolic continuous families such as $\alpha \in [0, 1]$, and explicit scenario execution at selected feasible states. General nonlinear partial identification, automatic model discovery and exhaustive enumeration of all economic histories are research extensions, not promises made by the core language.

6.7 A solver formulation for ordinary rounded disclosures

The debt example used an exact principal. Real disclosures often do not. Suppose debt is shown as 100 and interest as 5, each rounded to the nearest unit with the declared half-open intervals $D \in [99.5, 100.5)$ and $I_0 \in [4.5, 5.5)$. Using an unknown fraction α and unknown total D would introduce the product αD. That expression is bilinear and is outside the promised linear fragment.

Instead use principal amounts F and V. The compatible family satisfies

$$F, V \geq 0, \quad 99.5 \leq F + V < 100.5, \quad 4.5 \leq 0.05(F + V) < 5.5. \tag{6.11}$$

Future interest is $I_1 = 0.05F + 0.07V$. Its infimum is $0.05 \times 99.5 = 4.975$, attained with $F = 99.5, V = 0$. Its supremum is $0.07 \times 100.5 = 7.035$, approached as $F = 0$ and V tends to 100.5 from below, but not attained. A closed interval is a valid enclosure, but its upper endpoint must not be reported as attainable. The exact solver check separately tests that no better value exists and whether each endpoint is attainable.

For $D > 0$, the amount representation is equivalent to the fraction representation through $D = F + V$ and $\alpha = F/D$. At zero principal the fraction is undefined and must be omitted or treated as irrelevant, not recovered. A known fraction restriction $\alpha \geq a$ becomes $F \geq aD$ when $D > 0$ and a is fixed. If both a and D are unknown, that product remains nonlinear. Reformulation has a domain of validity; it is not permission to send arbitrary formulas to a linear solver.

The supported exact fragment contains rational linear equalities, weak and strict inequalities, and finite Boolean or enumeration choices. Guard partitions become finite disjunctions. Posting in integer cents introduces a different integer problem if performed before bounding. Consequently the request specifies whether bounds apply before quantisation, after a monotone declared rounding map, or to a bounded integer model. Approximate numerical valuations carry error metadata outside the exact rational kernel. Solver timeout, unsupported theory and unbounded objective have distinct responses; none is relabelled as infeasibility.

6.8 Building the family with explicit templates

The reconstruction input is a model-scope record, not an invitation to invent unlimited journals. It lists event templates, slots per entity and period, allowed contract classes, source mappings, opening-state domains, calendars, resolution and excluded operations. A template gives an incidence vector for postings and a transition for economic state. Northstar's annual bridges arise by substituting its collection, purchase and payment templates into opening-plus-movement identities.

For each allowed slot, create an activity flag and nonnegative amount; an inactive slot has zero amount. Instantiate the template's posting and state constraints, unify variables referring to the same observation or economic contract, and match report expressions to source domains through validated context mappings. Add cross-note restrictions and choice-group dependencies to this same formula. No separate account-by-account reconstruction is allowed. Mark every constraint with its evidence, rule or assumption origin.

Enumerate the finite structural choices in a deterministic order. For each cell, ask for feasibility and retain the continuous formula, rather than enumerating real-valued solutions. An infeasible cell preserves a sufficient conflicting subset; an undecided cell preserves its formula and reason. Search is complete within the declared scope only when every finite cell has been decided, every applicable observation has been matched or explicitly qualified, and no resource truncation remains. A witness establishes existence in one cell. It does not establish complete search, and complete search does not establish that the scope includes the company's real history.

In Northstar the declared templates identify annual collections, purchases and supplier payments, but retain every $F \in [0, 100]$. Adding a verified note that $F = 60$ narrows this family by conjunction. Adding a proposed "all floating" reading to that revised view produces a rejected cell with the conflicting note attached. Adding a rule that expenses a capital contribution would instead fail treatment admission: inconsistent evidence and unsupported accounting have different diagnoses.

6.9 Which details may safely be combined

A resolution record names the outputs, horizon, scenario family, grouping key, required state, transformation and exact or approximate status. Safe exact transformations are deliberately few. Cash balances in the same book and currency can be added. Homogeneous inventory lots may be combined under a weighted-average policy if ownership, cost basis, impairment state and future costing treatment coincide. Debt principal can be added only within a group sharing currency, coupon/reset function, fixing and payment calendars, maturity, options, guarantees and measurement method. The transition and reporting maps must be additive on that group.

The last condition supplies a short proof obligation. If a transition obeys $T(x_1 + x_2, u) = T(x_1, u) + T(x_2, u)$ and reporting is additive for every admitted driver u, induction preserves equality after grouping. This does not apply to covenant thresholds, counterparty netting, nonlinear guarantees or lot-specific impairment. Two fixed loans with the same dates and rate can share an execution state under this test. The fixed/floating pair cannot; neither can two policies with different guarantees merely because their current reserves match.

A changed horizon, scenario, rule version or requested output invalidates the resolution approval and reruns the sufficiency check. Extending beyond a debt maturity requires schedule detail that an earlier short-horizon view could omit. Approximate cohorts retain the detailed source references and a stated error criterion tied to outputs. Unless that error is bounded or evaluated, the grouping remains an approximation awaiting evidence, not an exact quotient of the history family.

6.10 A finite probability model that can actually be evaluated

Optional probability weighting begins with disjoint economic hypotheses and a declared joint measure. As an illustration, assign fixed and floating hypotheses prior probabilities 0.6 and 0.4. The stress itself leaves these weights unchanged and gives mean interest $0.6(5) + 0.4(7) = 5.8$.

Suppose genuinely new observed evidence has likelihood 0.2 under fixed and 0.8 under floating. The unnormalised posterior masses are 0.12 and 0.32; normalisation by 0.44 gives 3/11 and 8/11. These likelihoods are illustrative inputs, not extraction-confidence scores.

The finite reference algorithm enumerates joint hypothesis and innovation states, evaluates their declared joint mass and any observation likelihood once, checks the nonzero normalising constant, and propagates the entire statement vector on each state. It sums masses only across disjoint events. Two overlapping labels such as “fixed share at least one half” and “all fixed” are query views and cannot become two independent mixture components. Totals and their components share an observation block so their evidence is not counted twice. Continuous inference needs an additional parameterisation, density or conditional measure and an assessed numerical method; the finite algorithm supplies no guarantee for those extensions.

If the floating branch has an unsupported valuation operation, its mass 0.4 remains unresolved. The run may report the fixed result with successful mass 0.6, but cannot report 5 as the unconditional mean by renormalising away the failure. A separately requested mean conditional on successful execution must show that conditioning event and its mass. For unweighted families, an unresolved branch blocks a complete envelope over the family. Economic default is retained as a model outcome with remaining obligations, rather than being treated as a computational error to remove.

Chapter 7

The accounting language

7.1 Scope of the first language

AccDSL Core 0 describes entities and books, accounts, observations, shared unknowns, constraints, alternative interpretations, contract instances, accounting rules and reporting mappings. It has pure expressions and a restricted posting effect. Economic forecasting algorithms can live in typed plugins; unrestricted numerical code does not need to be reimplemented in the DSL. The grammar and interface details in Appendix A are normative for the first parser.

The compiler has six ordered stages: parsing with source locations; name and module resolution; type and temporal-context checking; lowering to the IR; constraint and rule-applicability analysis; and execution-plan construction. The compiler never silently supplies an economic assumption to make a program compile. A program can be valid as a historical representation while requiring additional assumptions before a particular simulation can run.

The language's meaning is defined by its IR and evaluator, rather than by the surface spelling alone. Formatting changes cannot change a model's semantic hash. A changed source span or reviewer decision changes evidence provenance even if the numerical model remains the same. The system therefore maintains separate evidence, semantic-model and execution identifiers.

The complete grammar is supplied as `spec/core-0.lark`, and the exact record fields as `spec/core-ir.schema.json`. The companion `spec/core-contract.md` defines reference resolution, expression arity, type/context checks, operations, failures and identity. Structural schema validation alone does not establish those semantic properties. ASCII identifiers and UTF-8 strings are used in this revision; currency suffixes, dates and decimal literals have explicit lexical rules. The grammar is checked by parsing the complete demonstration file, not by claiming that its parser also implements a compiler or solver.

7.2 Types that protect meaning

Type	Meaning and checks
`Money<C>`	Exact amount in currency C. Addition requires identical currency; conversion requires a dated rate and policy.
`Quantity<U>`	Units of shares, goods or other commodities. Quantities cannot be added to money or used as a functional-currency posting.

Type	Meaning and checks
`Rate<B>` and `YearFraction`	Rate with basis B, plus a separately computed accrual fraction. Simple and compound accrual functions are distinct.
`Instant`, `Period`, `Calendar`	Instant observations and flows over intervals; interval operations preserve calendar conventions.
`Account<K,Book>`	Account role K and owning book. Cross-book effects require separate balanced batches linked by an economic event.
`Observation<T>`	Source assertion with contextual and precision metadata; accessing its value retains its constraint and provenance.
`Symbol<T,D>`	Shared unknown with domain D; repeated references denote the same value in a history.
`Choice<E>`	A finite structural choice over enumeration E. No probability is implied.
`Distribution<T>`	Explicit probability measure with parameter, conditioning and measure metadata. It is not a Symbol or a Choice.
`SourceRef`, `RuleRef`, `ModelRef`	Versioned evidence or implementation references; unresolved references prevent concrete execution.
`Measurement<T>`	Value plus valuation date, basis, method, assumptions and numerical status. Recognition remains a rule decision.

Core expressions use exact integers and rationals, Boolean logic, enumerations, dates, records, lists, comparisons and registered pure functions. A decimal literal is parsed exactly, not first converted through binary floating point. Numerical plugins can return approximate measurements, but must declare their error or numerical status and the rounding used when posting money.

Missingness is a tagged type with distinct values: `notDisclosed`, `notApplicable`, `unreadable`, `withheld`, and `unresolvedMeaning`. Zero is an ordinary known number. A missing datum can be replaced by a declared Symbol or assumption in a model branch; the source observation stays missing. An unknown is not necessarily random, and an estimate is not necessarily missing.

Addition of monetary observations also requires compatible entity, perimeter, time, dimensions and basis. These checks operate on contextual expressions in addition to the currency type. An analyst can deliberately aggregate dimensions or transform a basis using an explicit function. For example, `sumOver(segment, ...)` records the dimension being removed and requires the segment partition to be declared; it does not assume every disclosed segment total is additive.

7.3 A small model with a continuous interpretation family

The following source uses a synthetic observation whose full context is stored in the evidence registry. The example imports the demonstration debt package, which uses one-year simple accrual, no fees and immediate rate resets. It is not an IFRS rule package.

All dependencies are instantiated in `fixtures/demo-debt-package.json`: the source reading, balanced opening state, currency precision, contract state and dates, reset and interest operations, closing and report mappings, request and expected outputs. `fixtures/borrowing.acc` is the complete source, including the retained earnings account and an `opening` declaration. The shorter

listings below explain its declarations in stages. They do not replace that full source.

```
module BorrowingExample version "1";
use DemoDebt version "1";
source note7 = external("synthetic-note-7");
entity Acme functional USD;
book Local entity Acme basis DemoDebt version "1";
account Cash : Asset book Local;
account Debt : Liability book Local;
account InterestExpense : Expense book Local;

observation debt : Money<USD> instant @2025-12-31
  = 100 USD from note7;
symbol alpha : Real in [0, 1]
  because "The fixed-rate proportion is not disclosed";
```

The remaining declarations use this same unknown fraction in both loan components. A change to the fraction therefore changes the whole debt mix consistently.

```
let fixedPrincipal = alpha * value(debt);
let floatingPrincipal = (1 - alpha) * value(debt);
constraint split:
  fixedPrincipal + floatingPrincipal == value(debt);

contract loans : DebtMix book Local {
  principal = value(debt);
  fixed_fraction = alpha;
  fixed_rate = 0.05 per year;
  initial_floating_rate = 0.05 per year;
}
alternative allFixed { assume alpha == 1; }
alternative allFloating { assume alpha == 0; }
scenario rateShock {
  set market_rate = constant(0.07 per year);
}
```

The parent model retains every $\alpha \in [0, 1]$. The two named alternatives are useful conditional views of that family, not a declaration that the endpoints exhaust it. Requesting the parent family's exact bounds produces $[5, 7]$ for the next year's expense in this simple model. Requesting a conditional run of `allFixed` supplies $\alpha = 1$. Requesting a weighted mixture requires a separate weight model; the two labels do not create equal weights.

An observation declaration is a reference to evidence plus a proposed typed reading. The compiler checks that the registry context agrees with the declared period and type. If the evidence is rounded, `value(debt)` denotes a shared constrained value rather than a silently exact 100. This synthetic registry explicitly declares an exact value, which keeps the illustration deterministic.

7.4 Posting rules and explicit exceptions

An event type declares the information needed by a rule. The example below assumes an event representing interest accrued and paid on the same date. Separate accrual and settlement events are needed when those dates differ.

```
event InterestPaid(principal: Money<USD>,
                   rate: Rate<annual_simple>,
                   fraction: YearFraction) book Local;

rule recogniseInterest for InterestPaid as e
when e.principal >= 0 USD {
  let charge = roundMoney(simpleInterest(e.principal, e.rate, e.fraction),
                          currencyScale(USD), "half_even");
  require charge >= 0 USD;
  post {
    debit InterestExpense charge;
    credit Cash charge;
  }
}
```

The rule cannot perform file I/O, mutate another entity or sample a random number. It returns a proposed batch. The runtime verifies balance and all preconditions before committing state. Negative interest requires an explicit supported rule; it is not mishandled by this intentionally restricted example. Rounding is a declared demonstration convention, not a universal accounting default.

Rules can declare `overrides ruleName`. The resulting override graph must be acyclic. For a concrete event, the evaluator determines all applicable rules and removes any that are overridden by another applicable rule. Exactly one maximal rule must remain for a required recognition operation. An empty set of applicable rules produces `UnsupportedTreatment`; more than one maximal rule produces `RuleConflict`. File order and a hidden numeric priority never break ties.

If a guard depends on a symbolic value, the solver partitions the current feasible family into the guard and its negation, subject to the existing constraints. This is a logical partition, not a random draw. An incomplete solver returns `UnknownApplicability`, not a guessed rule. Statically provable disjointness or total coverage can be cached; otherwise the runtime performs the same checks on every concrete event.

7.5 Alternative readings, assumptions and updates

An alternative adds constraints and possibly a supported semantic mapping to its parent model. Its record contains a rationale, supporting spans, opposing evidence, author, creation time and scope. Child alternatives inherit all parent constraints; they cannot quietly discard an inconvenient disclosure. A deliberate source rejection is a separate review decision with an explicit reason and a new evidence view.

Two independent alternatives may coexist, while two mutually exclusive readings of the same text require an explicit choice group. The representation stores these relationships in a constraint graph. It does not take a Cartesian product blindly. For example, interpreting an instrument as a liability affects both its balance-sheet account and the treatment of its periodic payments; those choices must be linked through one shared treatment variable.

New evidence narrows an existing family by conjunction when it concerns the same definitions

and reporting basis. If it changes the basis or corrects an earlier assertion, the system creates a revised evidence view and records the relationship. Posterior reweighting, when enabled, is a separate operation with likelihood provenance. A branch that becomes inconsistent is retained as historically considered and rejected, with the evidence that rejected it.

7.6 Modules and stable intermediate representation

A package identifies its namespace, semantic version, dependencies, effective dates, jurisdiction, supported entity classes, recognition operations and measurement policies. Standard packages define allowable treatments; industry packages add economic objects and specialised rules; company packages map source concepts and declare evidenced elections. A company extension cannot override a mandatory standard rule merely because its package is loaded last. An override is valid only at a declared extension point and within the selected standard's policy domain.

The IR has separate node classes for evidence, contextual observations, expressions, domains, constraints, alternatives, contracts, economic events, measurement results, rules, journal batches, report mappings and runs. Each node has a stable identifier and explicit input references. No expression stores executable host-language text. Plugins are referenced by versioned identifiers with typed signatures. This permits a second implementation to execute the same supported program without evaluating arbitrary embedded code.

The IR serialisation version is independent of the DSL version and of any accounting-standard edition. Migrations preserve the original payload and record changed meanings. A syntax migration can be automatic when semantics are unchanged. A changed recognition policy requires a new model version, even if the source file still parses.

7.7 Errors are part of the interface

The compiler and runtime return typed diagnostics with a source span, affected entity and period, dependency chain and suggested local action. The principal diagnostic categories have distinct recovery paths:

Diagnostic	Meaning and next action
`TypeMismatch`, `ContextMismatch`	Resolve incompatible units, periods, entities or measurement bases; do not coerce implicitly.
`MissingEvidence`	Supply a source or label a model input as an assumption.
`InfeasibleHistory`	Inspect the conflicting evidence and model constraints; preserve the rejected candidate.
`RuleConflict`, `UnknownApplicability`	Resolve rule scope or preserve a justified conditional partition.
`UnsupportedTreatment`	Add a reviewed capability or leave the affected output incomplete.
`UnbalancedBatch`	Reject the atomic update and diagnose the posting rule.
`SolverIncomplete`, `NumericalFailure`	Preserve the target and inputs; repair or change the computational method.
`InsufficientSimulationState`	Add declared conditional assumptions or request a family of results.

Warnings can describe low extraction confidence or an approximate economic model, but they do not replace these errors. A required unknown input prevents a concrete unconditional result. The analyst can add a declared conditional assumption, request bounds, or leave that output incomplete. Every path has a different, explicit result type; there is no generic fallback to zero.

Chapter 8

Reconstruction and execution procedures

8.1 Ingest, reconcile and construct histories

The historical procedure starts with a filing manifest. It records documents, reporting dates, publication dates, revisions and expected statements and notes. Tagged filings are parsed through an XBRL adapter; PDF and HTML sources are segmented into text, tables, captions and references. The original bytes and extraction output remain immutable. Re-running an improved extractor creates a new extraction version with links to the same source.

For each table, the importer first resolves headers, units, signs, periods and dimensions. It then proposes concept mappings. A deterministic validator checks subtotal arithmetic, duplicate facts, comparative consistency and cross-note references. Every proposed repair creates a new reading. The system may suggest that an OCR value is 10 rather than 70, but only a reviewed correction or an explicit alternative allows that reading into a validated model.

Reconciliation proceeds from local totals to movement schedules. A debt bridge may contain opening debt, issuance, repayment, acquired debt, interest accretion, currency effects, fair-value changes and closing debt. Every component receives a movement type. An unexplained residual is stored as a residual with a source and period; it is not automatically classified as repayment or issuance. A model can introduce an unknown decomposition of that residual, subject to evidence and economic constraints.

The reconstruction engine then performs the following finite procedure within a declared model scope:

1. Build shared variables for observations, opening states, contract terms and supported movement amounts. Attach original rounding domains.
2. Add identities, roll-forwards, account restrictions, contractual relationships and cross-period links. Add each semantic reading as a condition on the same joint model.
3. Propagate types, units, intervals and simple equalities. Reject context mismatches before invoking a numerical solver.
4. Ask the solver for feasibility. On infeasibility, return a sufficient conflicting subset of constraints, its source spans and model assumptions. A solver's conflict core need not be the smallest possible explanation.
5. Retain finite alternatives and continuous solution families symbolically. Materialise ex-

ample histories only when needed for review or execution.

6. Check whether each requested simulation has sufficient economic state. Missing maturities, reset dates or cohort composition become declared assumptions or explicit unresolved inputs.

Unrestricted journal enumeration would be unbounded: arbitrary offsetting transactions can be inserted without changing reports. The model therefore declares allowed event types, time resolution, relevant accounts, lot/cohort resolution and limits on latent contract instances. These limits describe the represented family; they are not evidence that all real possibilities were searched. The compiler canonicalises syntactic duplicates and equivalences established by its supported rewrite rules. It does not decide arbitrary economic equivalence. Probability measures attach to declared economic hypotheses or parameters, never to a count of encodings; adding a duplicate representation therefore cannot create extra probability mass or coverage.

The first implementation uses symbolic linear bridges and finite structural choices. Nonlinear economic models can be evaluated on feasible states through plugins. A solver timeout preserves the unresolved family and returns `SolverIncomplete`; it does not prune the family as impossible. Search budgets belong to run configuration and are recorded with the results.

8.2 Condition a run without rewriting history

A run selects an evidence view and cutoff, a historical family, a forecast starting date, scenario definitions, accounting packages, management policies, output queries and numerical settings. A run may use a concrete history, an exact symbolic family, a finite set of conditional histories, or weighted draws from a declared distribution. The output records which mode was used.

Suppose no debt maturity is disclosed. An analyst may run a conditional case that assumes all debt matures after the forecast horizon. The run is valid under that assumption even though the historical representation remains incomplete. The interface must display the assumption next to the result and provide sensitivity cases when the assumption is material. It must not label the assumed maturity as a discovered historical fact.

Management behaviour normally uses only information available at the decision date. A perfect-foresight stress case is a separate, explicitly labelled policy. Scenarios should share the same underlying innovations when estimating paired differences where appropriate, with stable stream identifiers so branch order does not change the draws. Common random numbers improve the comparability of paired simulations; they do not imply that every branch has the same economic event counts or shock distribution.

8.3 Schedule and commit events

Each event carries an economic occurrence time, accrual interval when relevant, settlement time, entity, contract or cohort identifier, currency, inputs and provenance. Its ordering key consists of date, declared phase, dependency edges and a deterministic tie identifier. The reference phases are: opening and scheduled contractual changes; market observations and rate resets; business events and decisions; accrual and measurement; settlement; period closing and reporting. A contract can require a more specific dependency. These phases are an engineering starting point, not a universal description of market timing.

Two events may share a tie order only if their order is either specified or irrelevant to the result. If payment before remeasurement and payment after remeasurement differ, an unresolved

ordering is a model ambiguity or an error. The engine does not conceal it behind alphabetical event IDs.

```
execute_event(event, state, run_context):
    verify prerequisites and information availability
    resolve applicable measurement and recognition rules
    compute proposed economic changes and measurements
    compute proposed journal batch(es)
    validate types, contexts, balance, constraints and idempotency
    atomically commit state, journals and explanation edges
    return new state, emitted dependent events, diagnostics
```

The atomic unit includes the economic state and its accounting effects. If posting fails, neither state changes. A repeated event ID with identical inputs returns the existing result; the same ID with different inputs is a conflict. Consolidation runs after entity books reach the required date and stores its adjustments separately. A failure in one scenario branch cannot mutate another branch's state.

8.4 Funding, tax and other simultaneous relations

Some financial models have circular dependencies. Interest affects taxable profit; tax and investment affect funding needs; funding affects interest. The reference executor first tries to resolve these through explicit timing. For example, beginning-of-period debt can determine current interest, with new borrowing affecting the next accrual period. That is acceptable only when the model's timing assumption is stated and appropriate.

Where the economic model genuinely requires simultaneous decisions, the dependency graph identifies the strongly connected component and delegates it to a declared equation or optimisation solver. The solver must return a solution, residuals, admissibility checks and uniqueness or multiplicity status. A contraction iteration requires a justified convergence condition; an arbitrary fixed iteration count is not a mathematical solution. Multiple admissible equilibria become alternatives when material. No solution produces a model outcome such as financing failure, or an explicit numerical failure, depending on the cause.

Funding policy must include available facilities, limits, covenants, collateral and allowed capital actions. The program cannot issue unlimited debt to keep cash positive. An inability to meet payments may be a legitimate default outcome. Default ends or changes contractual and accounting processes under the relevant event rules; it is not quietly discarded as a bad simulation draw.

8.5 Worked example: identical history, different future journals

Return to the company in Chapter 1. Immediately after the historical interest payment and closing, cash is 95, debt is 100 and equity is -5. There are no other assets, liabilities or transactions. Next year's expense is 5 in the all-fixed branch and 7 in the all-floating branch under the rate shock. The generated entries and closing statements are:

Output	All fixed	All floating
Debit interest expense / credit cash	5	7
Closing cash	90	88
Closing debt	100	100
Current-year profit	−5	−7
Closing equity after profit transfer	−10	−12
Cash change	−5	−7

The balance-sheet identity holds in each branch: $90 = 100 - 10$ and $88 = 100 - 12$. The demonstration reporting profile classifies interest paid as operating cash flow. A different permitted reporting profile changes that classification, while preserving the cash movement and source event. This example proves the intended distinction and its arithmetic; it does not estimate the probability that either debt structure describes a real company.

8.6 Worked example: a bond has several relevant amounts

A two-period zero-coupon bond raises cash of 100 and promises 110.25 at maturity. Assume a 5% effective rate per period, no transaction costs, no default, and amortised-cost treatment under a demonstration rule. The effective-interest schedule follows directly from $100(1.05)^2 = 110.25$:

Date	Opening liability	Interest accretion	Closing liability
Issue	0	0	100
End of period 1	100	5	105
End of period 2, before payment	105	5.25	110.25
Redemption	110.25	0	0

Issuance debits cash and credits the liability by 100. Each accretion debits interest expense and credits the liability. Redemption debits the liability and credits cash by 110.25. Legal face amount, carrying amount and market fair value remain separate fields. A market-rate shock may change fair value without changing this contractual payment schedule or this demonstration amortised-cost path.

Cash-flow reporting uses the actual payment event and its economic components; the annual accretion is non-cash. The applicable reporting package decides how to classify the maturity payment, rather than inferring it solely from the liability debit. Fees, modifications, early redemption, embedded options, hedges and foreign currency require additional event and measurement rules.

Ordinary equity issuance similarly separates cash, share quantity and legal capital. Ten new shares at 12, with par value 1, produce cash of 120, share capital of 10 and premium of 110 under a simple par-value profile. Shares outstanding increase by ten units. Direct issuance costs of 3 reduce the specified equity account and cash under the chosen illustrative treatment. The accounting rule and the cost's eligibility must be explicit; this example does not prescribe every jurisdiction's treatment.

8.7 Reporting and explanations

The reporting engine produces the balance sheet, income statement, comprehensive income, cash-flow statement, changes in equity and note schedules when the necessary modules are

present. Each output carries its basis, perimeter, period, scenario and interpretation. A comparative table cannot combine an original prior-year figure and a restated current-year comparative without displaying that choice.

An explanation is a dependency graph with typed edges: extracted from, interpreted as, assumed by, constrained by, generated by, measured by, posted by, aggregated into and supersedes. A cash-flow line must lead back to payment events; a fair-value movement must lead back to valuation inputs; a consolidated amount must show eliminations. A provenance path establishes how a result was obtained. It does not by itself establish that the underlying economic model is correct.

8.8 The corporate reference package as operations

The Northstar example defines a closed, reproducible corporate package rather than a universal standard. Its events separate performance, invoicing and settlement even where the example makes the first two simultaneous. Every payload includes entity/book, currency, economic date, period, source or assumption IDs and an information-view ID. Amounts are exact before the declared posting quantisation. Quantities and money are checked separately.

Event	Preconditions and state change	Posting
Deliver/invoice	Goods owned and available; control transferred; quantity removed at selected cost; unconditional amount due added	Debit receivable, credit revenue; debit cost of sales, credit inventory
Collect	Receipt no greater than selected unpaid receivable; reduce that balance; excess must be separately classified	Debit cash, credit receivable
Receive goods	Accepted owned quantity at supported unit cost; create supplier obligation and inventory lot	Debit inventory, credit payable
Pay supplier	Identified obligation and sufficient available cash; reduce both	Debit payable, credit cash
Payroll	Service period and cash obligation evidenced or assumed	Debit payroll expense, credit cash; accrual variant uses a payable
Acquire plant	Control obtained, cost known, in-service date/life/residual recorded; add asset cohort	Debit plant cost, credit cash or payable
Depreciate	In-service asset with remaining depreciable amount; apply dated schedule, capped by that amount	Debit depreciation, credit accumulated depreciation
Interest	Contract, reset state and accrual fraction known; create interest obligation, then settle if cash available	Debit interest, credit interest payable; debit payable, credit cash
Current tax	Reference tax base positive; 20% teaching rule; apply prepaid tax before further payment	Debit tax expense; credit prepaid tax or payable; settlement separately affects cash

Event	Preconditions and state change	Posting
Dividend	Explicit authorised distribution, sufficient funds and permitted capital action	Debit retained earnings, credit dividend payable; then settle
Close	Required period operations completed; preserve original movements	Transfer temporary income/expense balances to retained earnings

The current example combines accrual and settlement when they share a date, but the separate obligation remains the general operation. Customer advances are debit cash/credit contract liability and become revenue only on the supported performance event. Inventory costing requires an admissible policy; negative inventory and collection beyond an identified claim are rejected. Non-cash acquisitions, write-offs, impairment and currency movements require their own classified templates. An unknown residual never enters one of these operations merely to force a reconciliation.

Reporting maps asset and contra-asset balances into the balance sheet and liability/equity balances with their opposite sign. Before closing, current profit is included once in equity; after closing it is already in retained earnings. Income uses period movements in income/expense accounts, and OCI uses its own designated accounts. Cash flow sums cash legs by economic operation: Northstar's collections, suppliers, payroll, interest and taxes are operating, plant payments investing, and debt/equity proceeds or repayments and dividends financing. Non-cash depreciation is excluded from direct cash flow and added back only in the indirect reconciliation. Note schedules use the same contracts, lots and classified movements as the statements.

8.9 Scenario meaning and information access

An `ExogenousStress` supplies a path or joint driver law while holding the historical evidence model fixed. `ObservationalConditioning` additionally supplies actual observations and a joint probability model; the finite method in Chapter 6 evaluates the conditional distribution when its normalising mass is positive. `StructuralIntervention` requires a named structural model and replaces designated equations, retaining its exogenous noise. An absent structural model gives `UnsupportedScenario`, not an intervention inferred from an accounting identity (Pearl, 2009, Section 3.2.1). The textual Core 0 `scenario` declaration is an exogenous stress only; the other modes require the explicit typed API records.

At each decision date the runtime constructs an information view containing eligible disclosures, committed state through that date, currently observed drivers and the declared forecasting model. Future realised innovations and future outcomes are inaccessible. A stress path may be announced in advance, but this is an explicit visibility policy. Otherwise management sees only its current and past values. A bank's allowance at the opening date may use a forecast distribution of defaults; it must not inspect the outer simulation's future realised default. Inner valuation randomness has a separate stream domain from outer realised outcomes.

Funding follows a declared waterfall: available cash, committed eligible facilities up to their limits, and separately authorised capital actions. Each action checks collateral, covenants and settlement timing. If an amount remains due, record an economic funding failure and the unpaid obligation. If required valuation or accounting is unsupported, stop the affected dependencies as incomplete. The branch manifest preserves the last committed state, completed outputs, unavailable outputs and reason. These distinctions let an analyst see a meaningful bad outcome without confusing it with broken computation.

Chapter 9

Standards, industries and financing instruments

9.1 A common core with distinct accounting books

The common core governs evidence, time, money, events, journals and provenance. Reporting standards govern recognition, measurement and presentation within that core. The design preserves three levels of meaning: economic terms such as a promise to pay cash; an accounting treatment under a particular standard; and a company's reported label or extension. These levels are connected by explicit rules and mappings, rather than collapsed into one universal chart of accounts.

A rule package includes a standards register with jurisdiction, issuer class, effective dates, transition provisions, permitted elections, paragraph references, implementation version and tests. The package selector uses the entity, reporting basis, period and adopted elections. It must distinguish a standard's mandatory effective date from permitted early adoption. For example, the IFRS Foundation's IFRS 18 overview states an effective date of 1 January 2027 with earlier application permitted (IFRS Foundation, 2026c). A 2026 historical report cannot be assigned that presentation regime solely because the latest installed package includes it.

IFRS, US GAAP and ASBE packages share economic contracts and events where appropriate, but produce separate accounting books. A reconciliation between books consists of explained adjustments with origins in differing treatments, not a renaming of accounts. The first implementation can preserve all three source bases even while only a selected subset has executable rules. Such a source is represented as observed information with unsupported simulation treatments, rather than being silently converted to the supported basis.

The official IFRS overviews for IFRS 9, IAS 32 and IFRS 17 establish the relevant scope of financial-instrument measurement, presentation and insurance contracts (IFRS Foundation, 2026d,a,b). They are orientation sources, not substitutes for reviewed paragraph-level implementations. Public retrieval of the selected Chinese Ministry of Finance page and SEC API documentation failed during this campaign; the source register records these access gaps. ASBE and US GAAP semantic modules therefore have explicit primary-text acquisition and accounting-review tasks before they can be marked executable and compliant.

9.2 Non-financial companies

The initial industry package covers sales, collections, purchases, payments, inventory movements, payroll, capital expenditure, depreciation and basic financing. Revenue recognition is a separate rule from invoicing and cash collection. A service contract may generate a performance event before an invoice, creating a different asset from an unconditional receivable. Customer advances create obligations before revenue. Returns, rebates and variable consideration need separate assumptions and subsequent adjustments.

Inventory state includes quantity, cost, location, ownership and lot or cohort. A costing method is selected within the reporting package's permissible domain. An aggregate stock can support a simple model only if its approximation is declared. A commodity-price shock can expose differences between old and new inventory that an average value suppresses. Fixed-asset state likewise requires acquisition dates, components, useful lives, residual values, depreciation method, tax basis and impairment history when those affect the requested forecast.

Industry extensions add economic mechanisms without bypassing the accounting core. Construction requires performance and completion states; extractive industries require reserves, depletion and restoration obligations; utilities may require rate-regulation treatment; software businesses may require contract liabilities and development-cost decisions. The module manifest states what is represented, what is executable, and what remains aggregated.

9.3 Debt and hybrid financing

The contract layer for a debt instrument stores issuer and holder perspective, currency, issue and settlement dates, issue proceeds and legal face amount. Its payment terms include coupon/index/spread, reset and fixing dates, day-count convention, payment calendar, amortisation schedule and maturity. Its rights and protections include seniority, collateral, covenants, options, conversion terms, guarantees and linked hedges. Missing terms are explicit unknowns. A single carrying amount cannot stand in for this record.

The accounting layer stores the classification decision and its evidence, initial measurement, carrying amount, accrued interest, effective-rate method, unamortised fees or discounts, fair-value disclosures, impairment or credit adjustments where applicable, and presentation mappings. Contractual interest, effective-interest expense, realised cash payments and fair-value changes are separate quantities. The zero-coupon example in Chapter 8 provides a minimal conformance case for this separation.

The lifecycle must support issuance, recognition, accrual, reset, coupon, principal repayment, modification, derecognition, default, recovery and conversion. A modification event first determines which accounting operation the selected standard requires; it does not always mean issuing a new loan. An embedded option can affect valuation and classification without being exercised. A derivative linked as a hedge requires both the economic contract and any applicable accounting designation; economic risk reduction alone does not establish hedge-accounting eligibility.

Hybrid instruments make interpretation particularly consequential. IAS 32's overview discusses liability/equity classification and the separation of a compound instrument (IFRS Foundation, 2026a). The executable classification must consider the full terms and the selected standard version. US GAAP or ASBE treatment must be implemented under its own rules rather than inferred from an IFRS classification. Where the terms are not disclosed sufficiently, alternative classifications can coexist only if each is actually compatible with the known terms and applicable rules.

9.4 Equity and ownership

Equity has both monetary accounts and security quantities. The security register records share class, issued and outstanding quantities, treasury holdings, par or stated value where applicable, voting rights, dividend rights, redemption features, options and conversion relationships. Monetary equity accounts include contributed capital, reserves, retained earnings and other components required by the selected basis. These records evolve together through issuance, repurchase, cancellation, split, dividend and conversion events.

A stock split changes quantities and per-share measures without necessarily changing aggregate monetary equity. A cash dividend affects distributions and cash rather than current operating expense under the corresponding ordinary equity treatment. Repurchases and employee share awards require their own rules. Earnings per share uses a dated quantity schedule and rules for potential ordinary shares; simply dividing profit by the closing share count is insufficient when shares change during the period.

Funding simulations must connect the quantity register to market assumptions. An equity issue at a scenario-dependent price changes cash raised, shares and dilution. The model must state whether it targets a fixed cash amount, a fixed share count or a capital ratio. These are different management decisions even when the historical statement would report the same aggregate proceeds.

9.5 Banks: accounting capital is not regulatory capital

A bank package needs instrument and cohort states for loans, securities, deposits, wholesale funding, repos, derivatives, collateral and commitments. For interest-rate analysis, currency and repricing buckets matter alongside remaining maturity. For credit analysis, exposure, default state, loss severity, recoveries, guarantees and their dependencies matter. For liquidity analysis, deposit behaviour, collateral eligibility, encumbrance and payment timing matter. These states can be represented at an explicitly chosen aggregation level; the public record rarely identifies every contract.

Loan origination can debit loans and credit a customer deposit in the bank's book. Subsequent transfer to another bank produces settlement flows. Treating all loan growth as an immediate cash outflow would therefore encode a specific funding assumption, not a universal banking identity. A system-wide model must also link counterparties and settlement assets; single-company simulation records external counterparties without claiming to close the whole economy.

An allowance calculation is a dated measurement based on forecast information. For a single horizon, let EAD be a fixed exposure at default, PD the probability of default, and LGD the mean loss fraction conditional on default. Ignoring discounting, the illustrative expected loss in currency units is

$$L = EAD \times PD \times LGD.$$

With exposure 100, default probability 0.02 and loss given default 0.4, it produces 0.8. Changing those inputs to 0.04 and 0.5 produces 2, so the allowance increase is 1.2 if there are no other movements. This calculation is a demonstration model. A standard-specific expected-credit-loss implementation must determine horizon, staging or other classification, discounting, forward-looking information, write-offs and purchased-credit-impaired or other special cases. The simple formula is not an IFRS 9 or US GAAP compliance implementation.

Regulatory capital is computed in a separate prudential layer with its own perimeter, eligibility, deductions, risk weights and transition provisions. Accounting equity feeds a reconciliation to

capital; it is not automatically Common Equity Tier 1. Liquidity and capital constraints influence the management policy, which in turn generates asset sales, funding or capital events. A breached ratio can be an economically meaningful scenario outcome. No hidden capital injection is permitted to repair it.

9.6 Insurance: cohorts and future service

An insurer requires policy cohorts or groups with issue dates, coverage, premiums, benefits, expenses, lapses, claims development, guarantees and reinsurance relationships. Asset portfolios need their own contracts and valuations. The liability state must distinguish expected future cash flows, discounting, risk adjustments or margins, unearned profit measures where applicable, incurred claims and movements between opening and closing states. The IFRS 17 overview's distinction between measuring future cash flows and recognising profit as service is provided motivates this separation (IFRS Foundation, 2026b).

The module API accepts policy/cohort state, economic assumptions, non-economic assumptions and information date. It returns a measurement vector, movement attribution and proposed recognition events. Experience differences, changes in assumptions, finance effects and service effects remain separately identifiable. Reinsurance held has its own record and measurement; it is not an automatic percentage offset to every insurance liability.

Scenarios can alter discount curves, mortality, morbidity, lapse rates, inflation, catastrophe losses and policyholder behaviour. Their joint dependence and cohort exposure must be declared. A portfolio total can match history while hiding very different guarantee and duration profiles. This makes alternative economic histories especially valuable, but also limits what public-data reconstruction can identify. A full insurance package needs actuarial and accounting review beyond the core release.

9.7 Capability contracts for sector packages

Package	Required state and operations	Minimum acceptance evidence
Core corporate	Working-capital bridges, lots/assets, ordinary debt and equity, period closing and cash-flow mappings	Exact synthetic cases, reviewed public filing reconstruction, all statement reconciliations.
Financial instruments	Terms, calendars, resets, options, measurement categories, modifications and hedges	Contract schedule comparisons, valuation checks, paragraph-based posting cases and transition cases.
Banking	Credit cohorts, deposits, collateral, liquidity, regulatory capital, management actions	Allowance roll-forwards, repricing and runoff stresses, capital reconciliation and default/funding-failure cases.
Insurance	Policy groups, future cash flows, service and finance components, reinsurance and asset interaction	Cohort movement reconciliation, assumption change cases, service recognition and reviewed actuarial calculations.
Multi-basis reporting	Versioned IFRS/US GAAP/ASBE rules, entity elections and transition treatment	Matched economic-event cases with explained basis differences and independent accounting review.

Every package declares capabilities at operation level. “Supports bonds” is too vague; “supports fixed-rate, single-currency, non-callable amortised-cost issuance and redemption under this reviewed rule set” is implementable and testable. Unimplemented operations retain their source and contract representation while producing an explicit capability error when execution requires them.

9.8 A matched inventory case across reporting profiles

Standards packages need operation-level decisions. For interchangeable goods, the inspected AASB 102 text incorporating IAS 2 permits FIFO or weighted average in paragraph 25 and explains both in paragraph 27 (Australian Accounting Standards Board, 2015). The reproduced ASBE 1 text specifies FIFO, weighted average or specific identification in Article 14; its official Ministry version could not be retrieved in this review, so that reproduction has provisional source status (Ministry of Finance of the People’s Republic of China, 2006). Caterpillar’s company-authored 2024 report states that it principally uses LIFO and supplies a FIFO comparison (Caterpillar Inc., 2024, Note 1.C). That is evidence of a disclosed US-GAAP company policy; full ASC 330 paragraph authority remains an acquisition task. It is not a rule that every US company must use LIFO.

Take the same purchases in three separate books: ten units at 5, followed by ten at 7, then delivery of twelve units. FIFO charges $10(5) + 2(7) = 64$ to cost of sales and leaves inventory 56. LIFO charges $10(7) + 2(5) = 80$ and leaves inventory 40. Periodic weighted average uses cost $120/20 = 6$, charges 72 and leaves 48. The physical deliveries and supplier cash flows agree; profit and carrying inventory differ by the selected costing rule. A concept mapping cannot remove that difference. The FIFO/average selector in an IAS 2 profile rejects LIFO

as an unsupported election under the inspected paragraph. The US company profile admits its evidenced LIFO policy while retaining its narrower authority. An ASBE package remains source-provisional until the attributed text and applicable amendments are officially verified.

Inventory impairment illustrates another required memory. Under inspected AASB 102 paragraphs 28–34, an item costing 56 written down to net realisable value 50 records expense 6. If the conditions reverse and net realisable value becomes 54, reversal is 4, capped by original cost; the previous write-down must therefore remain in state. The reproduced ASBE 1 Article 19 also addresses reversal within the earlier provision. This paragraph is not used to assert the US treatment, whose primary authority was unavailable. Source limitations constrain the treatment claim, while the separate-book method and exact costing calculations remain fully specified.

9.9 A compound bond exposes classification dependencies

Consider an issuer receiving 100 for a two-year bond paying annual coupons of 5 and principal 100 at maturity, with a holder option to convert that principal into ten ordinary shares. Assume its currency is the issuer's functional currency, no other options, fees or contingent settlements, and that the conversion feature qualifies as equity under the selected rule. A comparable liability without conversion yields 10%. The inspected IAS 32-equivalent text requires separate liability and equity components and measures equity as the residual (Australian Accounting Standards Board, 2022a, paragraphs 28–32, AG30–AG32).

The liability's present value is

$$L_0 = \frac{5}{1.1} + \frac{105}{1.1^2} = \frac{11050}{121} \approx 91.3223, \qquad E_0 = 100 - L_0 \approx 8.6777.$$

Issue debits cash 100 and credits these two components. Under the stated effective-interest schedule, the first coupon leaves liability $1.1L_0 - 5 = 1050/11 \approx 95.4545$; after the second coupon it is 100. Redemption debits liability and credits cash 100. Alternatively, conversion at maturity debits liability and credits equity 100 while issuing ten shares; the original equity component remains in equity. AG32 supports the absence of a gain or loss on that maturity conversion. Exact rational valuations are kept before the declared monetary rounding.

A variable-number share settlement, cash alternative or other clause can change classification. The contract record must therefore retain these terms; it cannot select liability/equity accounts independently of the coupon, conversion and settlement rules. Early repurchase and induced conversion invoke separate operations: the inspected AG33–AG35 text describes allocation and the induced-conversion consideration difference. They are not implemented by changing the maturity date of the simple example. US-GAAP and ASBE classification require their own dated authority; the IFRS-oriented split is never silently reused under another basis.

9.10 Control and an elimination book

The consolidation procedure selects the control graph at each date, aligns supported accounting policies and presentation currencies, combines controlled entity books, and posts acquisition, elimination and non-controlling-interest adjustments in a separate book. A control conclusion needs evidence and a rule version. Ownership percentage alone is insufficient. Unknown purchase allocation, currency translation or unmatched intragroup differences remain named unresolved adjustments rather than unexplained balancing entries.

For a simplified acquisition, assume control is established, all net assets are already at the relevant acquisition measurements, no tax or goodwill arises, and non-controlling interest is measured proportionately. A parent has cash 60, investment 40 and equity 100; an 80%-owned subsidiary has cash 30, plant 20 and equity 50. Consolidation debits subsidiary equity 50, credits investment 40 and credits non-controlling interest 10. Group assets are cash 90 plus plant 20, and group equity is 100 attributable to the parent plus 10 to other owners.

A subsequent intragroup loan of 10 shifts cash from parent to subsidiary, creates a receivable and payable, and is eliminated by debiting payable and crediting receivable 10. Group totals remain 110. An unmatched loan currency or settlement date would block that exact elimination. If the parent instead sells its entire interest for 44 immediately after the initial state and loses control, the group removes subsidiary cash 30, plant 20 and non-controlling interest 10, recognises cash consideration 44 and gain 4; remaining cash/equity is 104. More complex acquisitions and retained interests require additional measurement operations; this example fixes the structure and arithmetic without certifying a complete consolidation standard.

9.11 A bank's cash, allowance and capital in one path

The bank reference state contains settlement cash 30, gross loans 100, allowance 0.8, deposits 110 and equity 19.2. Originating a loan of 20 first debits loans and credits the borrower's deposit. A subsequent transfer of 15 to another bank debits deposits and credits settlement cash. Loan origination and external cash use are consequently distinct events. Receive loan interest 6 in cash and credit deposit interest 2 to customer balances. Remeasure the allowance to 2.4, posting an additional loss of 1.6. The resulting state is cash 21, gross loans 120, allowance 2.4, deposits 117 and equity 21.6. Net assets are $21 + 120 - 2.4 = 138.6 = 117 + 21.6$.

Now 25 of deposits runs off. The declared funding policy uses cash first, then a committed facility with limit 10. It draws 4 and settles 25, leaving cash zero, deposits 92 and wholesale debt 4. A runoff of 40 instead leaves 9 unpaid even after the maximum facility draw. The simulator records default and retains the unpaid deposit obligation. It does not assume unlimited central-bank or shareholder funding.

The prudential calculation is a separate, illustrative policy. Deduct 2 from accounting equity to obtain capital 19.6; apply a 75% risk weight to gross loans for risk-weighted assets 90. The resulting ratio is 19.6/90. Neither the deduction nor the weight is a Basel default. Actual prudential packages require jurisdiction, exposure class, eligibility, buffers and dated rules, with a reconciliation from the accounting book. The structure follows the need to model capital and liquidity jointly in Behn et al. (2019, Section 2), without transferring that paper's estimated preferences or shock parameters to this bank.

For a richer credit adapter, partition borrower cohorts by currency, vintage, credit state, remaining maturity, collateral and measurement category. At date t, supply scenario probabilities, conditional default-time probabilities, exposure paths, conditional loss severity and discount factors. Its expected loss calculation has the declared form

$$L_t = \sum_s p_s \sum_{k \in H} d_{t,k} \Pr(\tau = k \mid s, \mathcal{F}_t) \mathbb{E}[\mathrm{EAD}_k \mathrm{LGD}_k \mid \tau = k, s, \mathcal{F}_t].$$

The conditional product expectation retains dependence between exposure and severity. The horizon H concerns eligible default times; associated cash shortfalls may continue later. The rule package decides twelve-month versus lifetime default scope, credit-state transfers, interest basis and write-off recognition. Changes in allowance must be attributed to origination, derecognition, transfers, revised expectations, currency and write-offs. Contract and prudential

support can be specified now, while full IFRS 9/CECL/ ASBE classification and paragraph-level validation remain later obligations.

9.12 Insurance service and reinsurance in a complete cohort

The example uses one-year policies and a one-year proportionate treaty starting when those policies are recognised, with uniform service for each. Both groups qualify for premium allocation through the one-year conditions in paragraphs 53(b) and 69(b); paragraphs 62–62A determine treaty recognition. Assume no acquisition cash flows, investment components, significant financing or onerous remaining coverage. Claims and recoveries are expected within a year of the claims being incurred. The inspected AASB 17 text incorporating IFRS 17 specifies the movements in paragraphs 53–59, allocation in B126 and separate reinsurance requirements in 63–70A (Australian Accounting Standards Board, 2022b). Gross and ceded groups retain distinct identities.

Begin with capital and cash of 20. Receive premium 120 and recognise a remaining-coverage liability of 120. Pay reinsurance premium 24 and recognise a coverage asset of 24. After half the uniform service, recognise revenue 60 by releasing gross coverage and reinsurance expense 12 by releasing its coverage asset. Estimated incurred claims are 40, of which 30 is paid; the remaining-claim risk adjustment is assumed to be 2. Gross claims expense is therefore 42 and incurred-claims liability is 12. Assume the treaty reimburses half the gross expected claims, recovery cash of 15 has arrived, the risk transferred is valued at 1, and reinsurer non-performance has zero expected effect in this example. The reinsurance claims asset is $20 + 1 - 15 = 6$, and recovery income is 21.

Cash is $20 + 120 - 24 - 30 + 15 = 101$. Assets are cash 101, remaining reinsurance coverage 12 and claims recovery 6, totalling 119. Liabilities are remaining gross coverage 60 plus incurred claims 12, totalling 72. Profit is $60 - 42 - 12 + 21 = 27$ and equity is $20 + 27 = 47$. These balances reconcile without netting reinsurance against the insurance liability. If counterparty credit risk changes, the recovery measurement changes through its own inputs; it does not mechanically halve every gross liability or risk adjustment.

Longer-duration business adds a fulfilment-cash-flow vector, discount curve, risk adjustment, contractual service margin, loss component and coverage-unit schedule. Select portfolio and annual cohort, assess onerous grouping, project cash flows within contract boundaries, discount with the declared financial assumptions, measure non-financial risk separately, and attribute subsequent movements to new business, experience, assumptions, finance and service. Future-service changes adjust the supported service-margin operation subject to the loss rules; current/past-service effects enter their specified result components. Unsupported variable-fee, transition or loss-recovery operations must stop those dependencies explicitly. A PAA example does not implement these longer-duration methods by implication.

An actuarial reserve adapter also needs a concrete estimation contract. For cumulative claims $C_{i,k}$ by origin year i and development year k, the chain-ladder point method estimates each factor from matched observed cells, $\widehat{f}_k = \sum_i C_{i,k+1} / \sum_i C_{i,k}$, then projects the latest cumulative claim along the remaining factors. A triangular example with rows $(100, 150, 180)$, $(120, 180)$ and (140) gives factors 1.5 and 1.2, outstanding amounts 36 and 112 for the unfinished rows, and total 148. Mack (1993, Sections 2–3, equations (1)–(3)) makes explicit the conditional-mean and variance assumptions and independence across origin years; its aggregate-error discussion retains dependence from shared estimated factors. A single final-development observation cannot supply an empirical tail variance without an additional assumption. The adapter therefore returns the point projection and marks uncertainty estimation unavailable in this tiny example. It supplies neither a full predictive distribution nor an IFRS risk adjustment. Calendar effects, inflation, changing claims handling and tail selection require separate sensitivity and validation before real use.

Chapter 10

Implementation contracts

10.1 Reference implementation and component boundaries

The initial implementation uses Python for orchestration and a deterministic reference evaluator, exact decimal/rational arithmetic for booked amounts, SQLite for transactional metadata and versioned records, and columnar files for large scenario outputs. DuckDB is an optional analytical reader. These are convenience choices justified by inspectability and modest initial scale; they do not establish production performance. The IR and conformance tests allow a later compiled executor without changing accounting meaning.

```
accdsl/
  evidence/       documents, spans, filing contexts, extraction versions
  model/          typed IR, entities, books, symbols, contracts
  language/       grammar, parser, resolver, type checker, lowering
  constraints/    propagation, solver adapters, bounds and conflict cores
  accounting/     rule selection, posting validation, closing, consolidation
  economics/      contract and behavioural plugin interfaces
  execution/      scheduler, branch state, stochastic streams, run records
  reporting/      statements, notes, movement schedules, explanations
  storage/        versioned records, migrations, result readers
  interfaces/     Python API, CLI, structured import/export
  packages/       reviewed demonstration and accounting modules
```

The first service boundary is local and synchronous. A job abstraction can be added for extraction and simulation without requiring distributed execution. The model compiler returns an immutable execution plan; the executor receives that plan and run inputs. Neither should retrieve live market data implicitly. Input datasets and rule packages are selected and versioned before execution.

10.2 Persistent records

Relation	Key information and constraints
`document`	Hash, original path/URI, filing ID, entity, publication and availability times, media type, revision links. Original bytes are retained.
`source_span`	Document hash, page/element locator, bounding box, text/table representation, extraction run ID.

Relation	Key information and constraints
`observation`	Context ID, value or missingness tag, precision domain, source-span IDs, semantic-reading ID, review status.
`context`	Entity/perimeter, concept, basis, currency/unit, instant or interval, dimensions and taxonomy version. Contexts are immutable.
`assertion_version`	Logical assertion ID, payload hash, economic validity, ingestion time, public availability, superseded version and reason.
`model_node`	IR node ID, type, serialised payload, dependency IDs, package/model version; no executable arbitrary text.
`constraint`	Expression, scope, source/assumption references, supported solver fragment, applicable alternative condition.
`alternative`	Parent model, added conditions, rationale, evidence, coverage declaration and rejection history.
`economic_event`	Stable event ID, contract/cohort, dates, quantities, dependencies, scenario and branch.
`journal_batch`	Event ID, book/basis, posting date, rule version, balanced postings, measurement IDs and atomic commit status.
`run`	Evidence and model hashes, cutoff, scenarios, branch selection, weight model, solver/numerical settings, seeds and environment.
`report_cell`	Run, branch/scenario, context, amount or result family, dependency references and completeness status.

Persistent identifiers are opaque; display labels can change without changing identity. References use foreign keys or validated content-addressed links. The full logical key of a reported fact includes its dimensions, not merely company, label and year. Duplicate assertions are stored as distinct source claims until a documented duplicate-resolution operation relates them.

Version records are append-only. A derived transaction-time interval can be closed by the next version event without deleting the original payload. Validity intervals can overlap when assertions conflict. Such overlap is diagnosed and resolved by the evidence view, not prohibited by a database constraint that would erase one assertion. Scenario states use structural sharing of immutable historical nodes and branch-local changes.

Large simulation output is indexed by run, explanation, parameter draw, scenario, innovation path, entity, book, date and account. Reducing this to means across all dimensions would destroy the product's central feature. Online summaries may compress draws within a declared distribution, while retaining conditional summaries and representative replay identifiers. A retention policy that removes individual traces must state which explanations can still be reconstructed.

10.3 Public API and result types

The Python and command-line interfaces expose the same operations:

```
ingest(manifest, extractor_config) -> EvidenceVersion
compile(source_or_ir, evidence_view, package_lock) -> ModelPlan | Diagnostics
reconcile(plan, cutoff, solver_config) -> HistoryFamily | Diagnostics
condition(family, assumptions) -> ConditionalFamily | Infeasible
simulate(family, scenario_set, run_config) -> RunResult
render(run_id, report_request) -> ReportBundle
explain(node_id, evidence_view_or_run_id) -> ExplanationGraph
compare(run_a, run_b, alignment_spec) -> Comparison
```

`HistoryFamily` contains a model-scope declaration, shared variables, constraints, finite choices, feasibility status and solver-completeness status. It may contain witness histories without claiming that those witnesses enumerate the family. `RunResult` is a tagged union of concrete outputs, conditional samples, certified bounds, explored ranges and unresolved outputs. Probability-bearing results reference a probability-model record. Results without that reference cannot be presented as probabilities.

Comparison requires an alignment specification: entity/perimeter, dates, reporting basis, scenarios and shared innovations where applicable. Comparing IFRS and US GAAP books holds economic events fixed and changes accounting rules. Comparing management responses holds the specified scenario and initial information fixed and changes the behavioural model. These are different questions and must not be combined into an unexplained difference.

10.4 Plugin interfaces

Every plugin declares a version, supported types, input information dates, determinism, numerical measure and capabilities. A contract plugin implements `schedule`, `observe`, `transition` and `payoff`; it may return future conditional events. A behaviour plugin implements `decide` and declares which state and scenario information it uses. A measurement plugin implements `measure` and returns an amount or vector with a basis and numerical status. A reporting plugin maps classified journals and balances to output contexts.

The plugin response must include all inputs that materially affect results. An external calculator's global evaluation date, market-curve cache, calendar version or random seed cannot remain invisible. For example, QuantLib's floating coupon interface makes fixing-date conventions and pricing objects relevant inputs; the adapter records them rather than relying on process-global state. ACTUS risk-factor observations receive explicit dates and identifiers.

The constraint interface accepts a typed formula and a requested operation: feasibility, witness, enumeration of declared finite choices, or bound. It returns `sat`, `unsat` or `unknown`, plus evidence appropriate to the operation. Timeouts and unsupported theories return `unknown`. A witness proves existence within the formula, not completeness of a recovered historical family.

10.5 Human review and model-assisted extraction

The review interface should put the source passage beside the proposed observation, context and reconciliation impact. An analyst can accept a mapping, add an alternative, reject an extraction, define a symbolic unknown or request more evidence. These actions create structured changes with reasons. The useful unit of review is a coherent disclosure and its links, not an isolated number detached from its note.

A language model can identify candidate tables, draft concept mappings, propose contract terms and suggest alternative readings. Its output must reference source spans and a schema. Deterministic checks reject fabricated source references, invalid units, inconsistent periods and unbalanced postings. A model-generated interpretation that passes those checks remains a proposal until its evidentiary status is established. Token probabilities and self-reported confidence are never automatically used as weights over economic histories.

For routine local research use, a lightweight interface and immutable source versions are sufficient. Multi-user approval hierarchies, cryptographic attestations and deployment security systems are outside this initial design. They can be added for a concrete operational requirement without changing the scientific representation.

10.6 Failure recovery and reproducibility

Extraction jobs and simulation jobs write explicit progress records and atomic results. Restarting a failed document conversion reuses its document hash and creates either the same idempotent extraction result or a new version under changed settings. Restarting a simulation resumes only from a committed state whose model and input hashes match the run. Numerical failures retain the inputs needed for a focused diagnostic.

A run manifest records code revision when available, package lock, source hashes, cutoff rule, model hash, scenario data, priors or weights, seed/stream scheme, solver settings, decimal and rounding conventions, calendars, environment and result locations. Absence of a usable Git revision is recorded as unavailable with content hashes, rather than inventing a commit.

10.7 Review decisions have consequences for later results

A proposed reading starts in `Proposed`. Acceptance records a reviewer, date, reason, verified locator and context, together with the reconciliation result. Acceptance means that the stated layer has been reviewed: accepting an extraction of a company's sentence does not accept its accounting treatment. Competing defensible readings enter `AcceptedAlternative` under a choice group. Rejection preserves the source and reason. A correction creates a new reading; replacing an accepted reading requires an explicit `Supersede` action. A prior reading therefore remains reproducible without entering the latest model.

Suppose OCR reads Northstar's original 120 as 720. The source span supports 120; correcting extraction creates a new reading linked to the erroneous one. The later revenue correction from 120 to 110 is different: it changes the company's reported assertion. The original extraction remains faithful to its document. Finally, a reviewer may accept that two notes conflict without accepting either as a unique model constraint. The view then contains a conflict set requiring alternatives or further evidence.

A changed reading or mapping invalidates descendant constraints, families, measurements, reports and cached runs for a new view. Unaffected dependencies can be reused. Old frozen runs retain their original decisions and evidence IDs; they are marked stale for current use, not overwritten. A journal derived under the old view is never silently attached to a new source. The local workflow needs no multi-user approval hierarchy to enforce these rules.

10.8 Identity must survive harmless reordering

The reference serialisation uses UTF-8 JSON with NFC strings, sorted ASCII keys, no insignificant whitespace and no floating-point literals. Exact numbers are reduced rational numerator/denominator strings; zero is 0/1. Duplicate keys, NaN, infinities and unpaired Unicode surrogates are rejected. Arrays preserve order unless the schema declares a set, such as a constraint conjunction or source-ID set, in which case IDs are sorted. This normalisation recognises representational equality, not arbitrary economic equivalence.

SHA-256 with a domain/version prefix identifies three different closures. Evidence identity includes source bytes, locations, extraction and review versions. Model identity includes typed IR, assumptions, rules, calculator locks and resolution, while excluding formatting and source byte offsets. Run identity adds the evidence view, horizon, scenarios, outputs, calendars, numerical policy, random-stream scheme and environment. A corrected source can change evidence and run identity without changing the numerical model; a new rate assumption changes the model as well.

An economic event ID derives from owner, schedule version, date, phase, operation and occurrence ordinal. The idempotency key pairs run ID and event ID. Identical payloads return the committed result; different payloads under the same key produce `EventIdentityConflict`. Independent events may reorder only when read/write sets are disjoint or a package proves commutation, including funding constraints. Alphabetical ordering cannot settle an economic timing ambiguity.

Stochastic keys contain seed, coupling group, entity, contract/cohort, economic date, shock kind and draw index. Branch enumeration order is absent. Shared coupling groups may use the same uniform draw with different marginal transforms. The reference fixture maps the first 64 bits of a SHA-256 key digest, interpreted as unsigned big-endian integer v, to $(v + 1/2)/2^{64}$. This fixes example replay; it does not validate a production Monte Carlo generator. Research runs select an assessed counter-based generator and record transforms and numerical environment. Inner measurement draws have different shock domains from outer realised defaults.

Schema migration preserves old and new payloads and a field/meaning mapping. A representation-only migration must demonstrate equivalent supported semantics; a changed recognition rule creates a new model. Restart resumes only a committed state whose locks and identities match. SQLite owns source, review and run metadata transactions; content-addressed files hold immutable bytes and JSON, while DuckDB reads versioned analytical exports. Each mutable transition has one owner.

10.9 Adapters exchange typed records

Exact method arguments and result fields are specified in `spec/core-contract.md`; `spec/core-ir.schema.json` supplies closed structural definitions. Every method returns a success record or a diagnostic with code, source IDs, entity/period, dependency path and retry status. The principal sequence is `ingest`, `review`, `compile`, `reconstruct`, `simulate`, `report`, with `bound`, `explain` and `replay` over frozen identities.

Arelle supplies candidate facts with QName, entity, context, period, dimensions, unit, accuracy, nil status and locator. Duplicates survive for review. Its archived `ModelInstanceObject` interface exposes these properties; validation success does not establish economic truth. Docling supplies page/table proposals, locations and conversion status. Missing headers and failed pages remain unresolved. Z3 receives named rational-linear assertions and finite choices; witnesses are checked against exact inputs and unknown results retain their reasons. The reference

checks exercise Z3; the other selected adapters remain interface designs grounded in inspected official source snapshots.

ACTUS receives contract terms, calendars and dated risk factors and returns schedules, transitions and payoffs. AccDSL applies accounting separately. QuantLib receives an explicit evaluation date, instrument, curves, fixings, calendar/day-count and measure; it returns dated measurements with numerical status. No process-global valuation date may leak between branches. Beancount exports concrete journals for later differential checks; it does not own symbolic families. Import failures retain request ID and source hash so retries are idempotent. Licence/version evidence remains in the software ledger; integration acceptance requires pinned releases or source digests.

Chapter 11

Acceptance and implementation sequence

The reference cases now make several of the acceptance obligations executable as design checks. Northstar provides full statements and a known compatible history; Apple supplies real source contexts and a bounded marginal stress; bank, insurance, consolidation and inventory fixtures test the core's broader state requirements. They do not constitute a production implementation.

11.1 Three different questions require different evidence

Engineering correctness asks whether the implementation follows the specified types, transitions and arithmetic. Accounting validity asks whether the selected rules faithfully implement the applicable standards and disclosed policies. Scientific validity asks whether the economic assumptions and uncertainty representation support the intended forecasts or comparisons. A balanced journal answers only part of the first question.

The implementation must therefore maintain separate validation results. Compilation and deterministic examples cannot establish forecasting quality. Agreement with a public report cannot identify an undisclosed contract term. A lower extraction error on a document benchmark cannot establish that the engine preserved every materially different scenario response.

11.2 Core conformance cases

Test	Required case	Pass condition
T01	Exact debit/credit batches, including reversals and period closing	Every committed book conserves signed balances; imbalance rejected atomically.
T02	Money, quantities, currencies, instants and interval flows	Invalid additions and context combinations rejected; explicit conversions traced.

Test	Required case	Pass condition
T03	Original filing, later correction and as-known cutoff	Earlier query reproduces the original evidence view; later view shows revision.
T04	Annual and interim cumulative flows with revised comparatives	Quarter derivation only across compatible contexts and vintages.
T05	Fixed/floating borrowing with identical historical postings	Both explanations survive; next-year expense is 5 and 7 under the stated shock.
T06	Continuous fixed share $\alpha \in [0, 1]$	Exact model returns $I = 7 - 2\alpha$ and extrema 5, 7 without assigning probabilities.
T07	Duplicated encoding of one history	Canonical duplicates preserve results and declared weights; no extra probability mass from an additional encoding.
T08	Linked interpretation across debt classification and payment treatment	Impossible combinations rejected through shared constraints.
T09	Rounded totals, duplicate facts and contradictory notes	Correct interval compatibility; no automatic zero or balancing plug.
T10	Infeasible or unsupported constraint problem	Conflicting sources exposed; timeout/unsupported case reported as unknown.
T11	Zero-coupon bond and ordinary equity examples	Carrying amounts, payments, share quantities and report mappings match derivations.
T12	Foreign-currency asset, translation and cash movement	No mixed-unit balance test; translation effects separated from cash flows.
T13	Acquisition and intragroup balance elimination	Perimeter and adjustment book reproduce group totals without inventing capex.
T14	Conflicting recognition rules and unknown guards	Explicit conflict or symbolic partition; file order has no effect.
T15	Insufficient cash with constrained funding	Funding failure/default retained as an outcome; no hidden borrowing.
T16	Expected-loss measurement before future default is realised	Measurement uses only information allowed at that date.

Test	Required case	Pass condition
T17	Repeated/reordered independent events and interrupted execution	Idempotent replay; branch isolation; identical committed outputs.
T18	Conditional distributions and optional weighted mixture	Branch distributions retained; mixture agrees with declared weights and total probability; unweighted family never labelled probabilistic.

For synthetic cases, expected histories, excluded histories and scenario consequences are known by construction. Tests should include counterexamples that balance but have the wrong economic effect, such as treating a non-cash acquisition movement as operating cash. Such tests discriminate between arithmetic validity and the intended semantics.

Differential tests against an existing ledger check only the shared supported subset. ACTUS schedule comparisons check contractual events, and QuantLib comparisons check selected valuation calculations under matched conventions. Disagreements must be classified as different conventions, unsupported cases or implementation errors before one tool is treated as an oracle.

11.3 Extraction and reconstruction evaluation

A company corpus must include different standards, industries, languages, filing formats, extensions, interim calendars and restatements. Company and time splits are required so repeated report templates or revised versions do not leak into evaluation. Public benchmark tables are useful development material, while acceptance uses complete disclosure packages that include notes and their relationships.

Reviewers annotate source facts, contexts, mappings, required relationships, known ambiguities and unsupported assertions. Relevant measures include correct source-linked values and contexts, unresolved extraction volume, coverage of note obligations, reconciliation failures and invented facts. Report each separately. A single overall score can hide a serious error in a small but consequential financing note.

Historical reconstruction should test whether the known synthetic history remains feasible, whether known incompatible histories are rejected, and whether economically distinct alternatives are preserved. On real companies, public data generally cannot establish the true journal. Review instead assesses evidential compatibility, unsupported commitments and conditional economic consequences. Later disclosures may provide discriminating evidence for previously ambiguous terms.

No universal numerical threshold for semantic coverage or extraction accuracy is fixed in this document. Before a release evaluation, an accounting-reviewed test corpus and task-specific acceptance thresholds must be recorded in an evidence contract. Hard conformance failures, fabricated citations, silent discarding of valid alternatives and undocumented accounting adjustments veto acceptance regardless of an aggregate score.

11.4 Evaluate the DSL against the strongest practical baseline

The comparison uses the same IR, accounting examples and tasks. One group of models is authored through the typed Python API; another through the DSL. Both receive equivalent documentation and examples. Tasks include encoding a movement schedule, adding an alternative interpretation, changing an effective date and tracing a simulated amount. Measure correctness, material omissions, time to a reviewed result, ease of explaining a change and maintenance effort.

This comparison tests whether the notation and compiler help users; it is not a comparison between careful DSL code and unstructured spreadsheets. If the DSL offers little advantage, retain the shared IR and library and narrow the textual language. Small pilots nominate improvements. A claim of superiority requires a predeclared comparison and uncertainty evidence, not a few favourable demonstrations.

11.5 Forecasting evaluation comes later

Forecast evaluation uses historical information cutoffs, fixed data vintages, declared economic scenarios and proper separation of training and testing periods. Baselines should include a naive persistence or ratio model, a tuned conventional financial model, the structural model with a single selected interpretation, and the same model preserving alternatives. This identifies whether any benefit comes from better economics, more information or the representation of ambiguity.

Construct the inexpensive comparators before choosing a complex forecasting method. A no-growth operating forecast with scheduled contractual financing tests whether the model adds value beyond existing obligations. A constant margin and turnover model tests whether detailed operations improve on familiar financial ratios. A trailing-average growth model tests whether fitted dynamics improve on recent experience. A fixed-exposure contractual stress model tests whether estimated management responses help beyond repricing the existing book. Each comparator must generate coherent accounts through the same reporting rules and use the same historical information. In synthetic cases, the known hidden state also supplies an oracle conditional on the stipulated scenario.

Evaluate these comparisons separately under stable conditions, large rate resets, refinancing pressure, credit deterioration and changes in reporting perimeter. A pooled average can conceal failure in the situation the product is intended to explain. Supported underperformance against a cheap comparator in a declared critical situation vetoes promotion of the complex forecasting method and must lead the result discussion. These checks are falsification tests, not training targets or evidence of superiority when merely passed.

Evaluate point forecasts where they are a stated target, distributional calibration when probabilities are justified, and decision-relevant robustness when only feasible sets are available. A wide feasible set is not automatically a good prediction, and an unweighted set cannot be assessed as if it were a 95% predictive interval. In multi-scenario stress tests, scenario severity and probability remain separate inputs.

Any serious numerical comparison must follow the repository's research plan, assumption audit, baseline, veto, uncertainty and result-record policies. The present document runs only deterministic checks of its own examples. It neither selects a scientific forecasting default nor reports comparative performance.

11.6 Implementation milestones and dependency order

Stage	Concrete implementation deliverable	Exit evidence
P0	Evidence registry, contextual types, versioned IR and exact posting engine; Python constructors for the running cases	T01–T04, T09 and T17; source inspection and replay.
P1	Finite choices, shared symbols, linear solver adapter, historical bridges, conditional simulation and report explanations	T05–T10, T14 and T18; synthetic family preserved.
P2	Textual Core 0 frontend with identical IR; initial corporate package; document/XBRL import adapters	Parser/API equivalence; T11–T13; reviewed three-date public-company case.
P3	Contract and valuation adapters, dated standard packages, richer financing and tax/funding policies	Contract/valuation differential cases; accounting review; T15 and transition cases.
P4	Bank package, allowance models, regulatory reconciliation and management response alternatives	T16; banking-specific reviewed histories and stresses.
P5	Insurance cohort package, reinsurance and actuarial measurement adapters	Reviewed cohort roll-forwards, information-date and service/finance tests.
P6	Optional probabilistic inference, model-weight sensitivity and broader scenario research	Explicit targets, sampler/numerical validity, prior sensitivity and downstream evaluation.

These stages specify dependency order, not dates or staffing estimates. Banking and insurance representation requirements influence P0 types, although their measurement engines arrive later. Probability weighting is optional; P1 already produces useful conditional cases and bounds. A failure of a chosen extractor or inference algorithm triggers a local replacement or repair, rather than abandoning the product's representation requirements.

11.7 Decisions and replaceable assumptions

The requirements fixed by this design are preservation of evidence and alternatives, explicit time and basis, balanced and admissible posting, economic state beyond journals, and distinct set/probability semantics. Replaceable choices include the storage engines, parser generator, solver, numerical libraries, aggregation resolution and user interface. Every scientifically material aggregation, behavioural parameter, prior or truncation rule must record provenance, purpose, failure mode and a diagnostic before it becomes a default.

The immediate implementation task is P0 using the complete Northstar and debt fixtures, the temporal/review cases and the declared IR. Broader corporate and sector support grows through the specified operations and reviewed examples; no generic rule package is presumed to cover all financial statements.

11.8 Concrete conformance and corpus protocol

The fixture directory contains independently entered inputs and expected outputs for Northstar, rounded borrowing, evidence views and sector cases. `check_repair_conformance.py` recomputes exact journals and statements, constructs the joint linear bridges in Z3, checks strict bounds, and exercises replay, atomic rejection, finite Bayes updating and funding failure. `check_core_contract.py` checks grammar/schema examples, canonical identities, stable streams, information restrictions and source inventory. These scripts are reference oracles for portions of the design. They do not test an unimplemented full DSL compiler, all adapters or standards compliance.

Every conformance case must specify input files, the operation requested, expected values or typed errors, comparison convention and supported scope. Exact money, finite probabilities, linear witnesses and example journals compare as rationals. Approximate valuations require an error bound or a justified tolerance tied to the valuation scale. A missing expected result cannot become a passing test simply because the operation returned something. An implementation release must pass all applicable positive and negative fixtures and preserve unresolved operations; the arithmetic oracle alone does not release it.

The disclosure corpus grows in distinct strata: an ordinary US-GAAP issuer (Apple, with Caterpillar as a contrasting inventory policy), an IFRS corporate issuer, an ASBE corporate issuer, a bank and an insurer with contract-group disclosures. The acquired Apple package is the initial real stratum, not a representative population sample. Each later stratum requires an annual and interim package, at least three dates, policies, note links and a changed comparative or a documented absence of one. Synthetic restatement cases supply controlled revisions when the real package contains none. Candidate issuer selection, acquisition status and operations to annotate are explicit in `fixtures/corpus-protocol.json`; unacquired strata remain pending.

Annotators first record literal/context readings without seeing the model's predictions, then mark source conflicts, mapping relationships, narrative qualifications and admissible alternatives. A second accounting-aware reviewer adjudicates material disagreements while preserving both initial annotations. Reports from the same issuer family and duplicated tables cannot be split between development and heldout evaluation. Forecast tests freeze the actual publication cutoff; later revisions are separately labelled outcomes and may not enter predictor inputs. Extractor upgrades rerun the same frozen corpus.

For the initial supported release, loss of a source span, wrong entity/unit/ period/basis, invented evidence, hidden financing, dropped feasible branch, incorrect cash reconciliation or unreported unsupported treatment blocks acceptance. All selected critical facts and operations must have exact reviewed answers or the expected unresolved response. These zero-tolerance conditions follow from the product's meaning-preservation requirements. Non-critical extraction quality is reported by stratum and error category; no arbitrary overall accuracy threshold certifies whole-report understanding. An extraction benchmark needs its own sample size and uncertainty protocol before making statistical accuracy claims.

The DSL and typed-library comparison uses identical packages, evidence, schemas and runtime. Users perform the same tasks: correct a context, add a shared uncertainty, represent a restatement, add a supported event and explain a forecast difference. Task order is counterbalanced and experience recorded. Both frontends receive the same validation and error messages. Measure successful completion, material semantic errors, time and maintenance effort with uncertainty appropriate to the participants; report no winner from the current syntax checks. A textual language earns its maintenance cost only if that comparison supports its use.

11.9 Workload limits and release evidence

The first integration workload is one local company package, up to 10,000 observations, 100 declared aggregate event slots per period, 64 materialised finite-choice cells and a 30-second budget per solver operation. These are provisional engineering limits for bounded failure behaviour, not measured capacity or scientific defaults. They must be configurable and recorded. Reaching a limit yields partial coverage with the unvisited formula/cells, never an exhaustive-history claim. Peak memory and latency are measured only after an implementation exists; a slower but semantically correct result does not become a scientific failure.

Engineering release needs compiler/API equivalence, transactional replay and adapter subset checks. Accounting release additionally needs paragraph-level authority, dated applicability, exact fixtures and independent domain review for every promoted operation. Forecast promotion additionally needs preserved information sets, the stated baseline ladder, statistical uncertainty and critical-regime vetoes. Human comprehension review remains separate from all three. The present repair supplies a richer design candidate and reference checks while preserving these outstanding acceptance decisions.

Appendix A

Core grammar, records and execution details

A.1 Core 0 grammar

The normative grammar is reproduced from `spec/core-0.lark`. Names are case-sensitive ASCII identifiers; strings are UTF-8 with JSON escaping. All literal words in the grammar are reserved. The companion contract specifies valid dates, exact rational literals, argument arity, type/context checks and package resolution. A syntax parse is not semantic validation.

```
// Normative Core 0 syntax, Lark EBNF. UTF-8 input; ASCII identifiers.
// Longest token wins; keywords cannot be declaration identifiers.
start: "module" NAME "version" STRING ";" declaration*
?declaration: use | source | entity | book | account | observation | symbol
            | choice | binding | constraint | alternative | contract | event
            | rule | scenario | report | opening
use: "use" NAME "version" STRING ";"
source: "source" NAME "=" expr ";"
entity: "entity" NAME "functional" NAME ";"
book: "book" NAME "entity" NAME "basis" NAME "version" STRING ";"
account: "account" NAME ":" NAME "book" NAME ";"
observation: "observation" NAME ":" type time_spec "=" expr "from" NAME ";"
time_spec: "instant" DATE | "period" DATE "to" DATE
symbol: "symbol" NAME ":" type "in" domain "because" STRING ";"
domain: "[" expr "," expr "]" | "{" expr ("," expr)* "}"
choice: "choice" NAME "in" "{" NAME ("," NAME)* "}" "because" STRING ";"
binding: "let" NAME [":" type] "=" expr ";"
constraint: "constraint" NAME ":" expr ";"
alternative: "alternative" NAME "{" assumption* "}"
assumption: "assume" expr ";"
contract: "contract" NAME ":" NAME "book" NAME "{" field* "}"
field: NAME "=" expr ";"
event: "event" NAME "(" [parameters] ")" "book" NAME ";"
parameters: NAME ":" type ("," NAME ":" type)*
rule: "rule" NAME "for" NAME "as" NAME ["overrides" NAME ("," NAME)*] "when" expr "{" rule_statement* "}"
?rule_statement: binding | require | post
require: "require" expr ";"
post: "post" "{" posting+ "}"
posting: SIDE NAME expr ";"
scenario: "scenario" NAME "{" setting* "}"
setting: "set" NAME "=" expr ";"
report: "report" NAME "for" NAME "{" field* "}"
opening: "opening" NAME "for" NAME "from" expr ";"
type: NAME ["<" type ("," type)* ">"]
?expr: disjunction
?disjunction: conjunction (OR conjunction)*
?conjunction: comparison (AND comparison)*
?comparison: sum [COMPARE sum]
?sum: product (ADD product)*
?product: unary (MUL unary)*
```

```
?unary: UNARY unary | postfix
?postfix: atom ("." NAME)*
?atom: number | STRING | DATE | BOOL | call | NAME | "(" expr ")" | array | record
call: NAME "(" [expr ("," expr)*] ")"
number: NUMBER [NAME | "per" "year"]
array: "[" [expr ("," expr)*] "]"
record: "{" [record_field ("," record_field)*] "}"
record_field: NAME ":" expr
SIDE: "debit" | "credit"
OR: "or"
AND: "and"
UNARY: "-" | "not"
COMPARE: "==" | "!=" | "<=" | ">=" | "<" | ">"
ADD: "+" | "-"
MUL: "*" | "/"
BOOL.2: "true" | "false"
DATE: /@[0-9]{4}-[0-9]{2}-[0-9]{2}/
NAME: /[A-Za-z_][A-Za-z0-9_]*/
NUMBER: /(?:0|[1-9][0-9]*)(?:\.[0-9]+)?/
STRING: /"(?:[^"\\\x00-\x1f]|\\(?:["\\/bfnrt]|u[0-9a-fA-F]{4}))*"/
%import common.WS
%ignore WS
%ignore /\/\/[^\n]*/
```

A.2 Required core record definitions

The closed machine-readable definitions are in `spec/core-ir.schema.json`. They define contexts, source spans, observations, narratives, coverage units, mappings, partitions, entity/control records, books, accounts, symbols, choices, expressions, constraints, assumptions, alternatives, contracts, events, rules, packages, measurements, states, journals, scenarios, information views, review decisions, evidence views, model scopes, historical families, probability models, report mappings, result cells, branches, runs and diagnostics. The schema is normative for field spelling and structural shape; the semantic contract is normative for reference resolution, units, contexts, applicability and operations.

For example, these selected records reproduce the schema's field names. Each field is required; a nullable link is represented by explicit null.

```
EvidenceView {
  id, cutoff, system_as_of, mode,
  documents[], observations[], decisions[],
  conflict_sets[][], transforms[]
}
Context {
  id, entity, perimeter, concept, basis, measurement,
  time, unit, currency, taxonomy_version, dimensions[]
}
Symbol {
  id, type, context, domain, uncertainty_kind,
  scope, dependencies[], admission, provenance[]
}
Journal {
  id, event, book, date, accrual_period,
  postings[], rule, measurements[], assumptions[],
  precondition_hash, explanation
}
ResultCell {
  id, context, history_condition, scenario, path, kind,
  value, expression, lower, upper,
  lower_attained, upper_attained, sample_ref,
  certificate, reason, dependencies[]
}
```

A value is a reduced rational, or a typed symbolic/domain reference. A posting is nonnegative with its side carrying the sign, in the book's functional currency. Original units and the valuation reference survive conversion. Monetary contexts require currency; a period flow requires a calendar and exclusive end boundary. Zero postings may preserve traceability but do not establish that an economic event occurred. The amount may have finer precision before the declared posting-rounding operation.

Schema validation rejects unknown fields but cannot prove that a treatment is permitted, a reference points to the correct kind of record, or two periods are comparable. Those semantic checks precede solver or runtime dispatch. The structural fixture and complete demonstration source/package are supplied separately so that a future implementation can check both layers explicitly. `fixtures/core-ir-demo.json` supplies an authored expected lowering; the reference checker evaluates its selected debt expressions. A general source-to-IR compiler and complete external-registry validation remain to be implemented. Domain records retain disclosed display text and rounding rules alongside their exact bounds.

A.3 Typing and partial-information rules

For expressions x and y in the same currency C, the money addition rule is

$$\frac{x : \mathrm{Money}\langle C\rangle \quad y : \mathrm{Money}\langle C\rangle}{x + y : \mathrm{Money}\langle C\rangle}.$$

Context compatibility is a separate premise for observations. There is no analogous implicit rule for different currencies. Explicit conversion uses a dated measurement and records its rate and purpose. A rate-times-principal calculation is typed through a declared accrual convention; the example's annual simple rate and year fraction produce money in the principal's currency.

Known expressions evaluate normally. Shared symbols evaluate to symbolic expressions subject to a joint constraint store. Missing source values cannot enter arithmetic until the model supplies a symbol or a conditional assumption. Boolean guards over symbols produce formulas, not the value false. This avoids the common error in which "unknown whether impaired" silently means "not impaired." Division requires a proven nonzero denominator or a partitioned domain that excludes zero; otherwise the expression remains unsupported for concrete execution.

Logical queries first require a nonempty feasible family. An empty family returns `Infeasible`; it must not produce vacuous answers to substantive questions. For a nonempty family, constraint evaluation has three results. `entailed` means every state in the declared feasible family satisfies the proposition. `refuted` means none does. `undetermined` means both remain feasible or the solver has not decided; the response distinguishes those causes. Probabilistic truth is a separate query requiring a measure. A language-level Boolean value must never be inferred from an arbitrary probability threshold.

A.4 Operational rule selection

Let $V(e, h)$ be the applicable rules for event e in concrete history h, after package/date selection and guard evaluation. Let $r \prec r'$ mean that r' validly overrides r through the declared acyclic graph. Define

$$\mathrm{Max}(V) = \{r \in V : \nexists r' \in V \text{ with } r \prec r'\}.$$

For one required recognition operation, a singleton selects that rule; the empty set yields `UnsupportedTreatment`; a larger set yields `RuleConflict`. Multiple distinct operations, such as measurement and recognition, are evaluated through separate declared interfaces. A rule that intentionally produces no postings must belong to a supported non-recognition event operation, whose economic state effect is still recorded.

Given a selected rule, the evaluator computes all local bindings and preconditions, emits proposed postings, verifies Equation (6.2), validates account/context restrictions and commits the event transaction. Failure returns the original state plus a diagnostic. This defines the atomicity requirement independently of the storage technology.

A.5 Example run request and response

The following request executes two named conditions under one scenario. It does not request an aggregate probability distribution.

```
{
  "model": "BorrowingExample@1",
  "evidence_view": "synthetic-2025-exact",
  "starting_state": "after-2025-close",
  "horizon": {"start": "2026-01-01", "end_exclusive": "2027-01-01"},
  "conditions": ["allFixed", "allFloating"],
  "scenarios": ["rateShock"],
  "mode": "conditional",
  "weight_model": null,
  "reports": ["balance_sheet", "income", "cash_flow", "equity", "journals"],
  "package_lock": "demonstration-packages-v1"
}
```

The response contains two separately labelled results with interest expense 5 and 7, cash 90 and 88, debt 100 in both, and equity -10 and -12. A request for the parent symbolic family returns the formula $7 - 2\alpha$ and, under exact linear bounds, $[5, 7]$ before monetary quantisation.

The API rejects an unconditional mean request without a weight model. A user can instead request an arbitrary clearly labelled representative case, which does not stand in for the feasible family.

A.6 Requirement traceability

Requirement	Primary implementation area	Conformance cases
R1	Evidence registry and extraction adapters	T03, T09
R2	Context/version model and cutoff queries	T03, T04
R3	Types, mappings and package selection	T02, T12, T13
R4	Symbols, choices and constraint graph	T05–T10, T14
R5	Posting, closing and consolidation	T01, T11–T13
R6	Contract and behavioural interfaces	T05, T11, T15, T16
R7	Conditional output and probability types	T06, T07, T18
R8	Explanation graph and report dependencies	T09, T11–T13
R9	Capabilities and versioned modules	T02, T14–T16
R10	Run records, scheduler and replay	T03, T17, T18

Appendix B

Research basis and outstanding review

B.1 Source collection and reading scope

The companion library contains 32 verified paper PDFs: 25 retrieved from external sources and seven preserved copies from the original collection. Filenames use the title, first author's surname and year. Publication years are used where an archived working version underlies a later publication; the catalogue identifies the actual version read. The original collection has not been renamed or modified. Standards pages, contract specifications, software source snapshots and metadata are stored separately from papers.

The survey considered accounting algebra and DSLs, corporate and bank simulation, incomplete and temporal databases, provenance, inverse problems, partial identification, model averaging, causal scenarios, contract languages, probabilistic programming, document extraction and financial reasoning. OpenAlex and Crossref supplied dated metadata, while primary publisher, institutional and author copies supplied technical support. Public-index results were checked against the downloaded title pages; HTTP success alone was not accepted as a valid paper download.

Technical reading was targeted to the mechanisms used in this design. The full PDF is archived even where only specified sections support the argument. No claim is made that every theorem, appendix or empirical result in every archived paper was independently replicated. Representative support anchors are listed below; the source-support ledger contains the per-source scope and qualifications.

Source	Inspected technical support	Design use
Nehmer and Robinson	Sections 2–5: account states, transactions and aggregation	Conservation plus admissibility.
Doshi	Chapters 4, 7–9 of the 2024 thesis	Dates, logs, corrections and DSL scope.
Shahnazarian	Sections 2.3–2.8; relevant Chapter 3 equations; data and evaluation sections; Appendix B	Accounting/behaviour separation; limits of data transfer.
Pierson	Sections 3–4 and reporting structures; supplementary Vensim equations	Accrual/payment bridges and reporting accumulators.

Source	Inspected technical support	Design use
Libkin	Sections 2–5: incomplete states, conditions and certain answers	Shared unknowns and joint alternatives.
Green et al.	Definitions 3.1–3.2; Section 4	Composable relational provenance; limits outside positive operations.
Stuart	Section 2.2, observation equation and prior/regularisation discussion	Inverse-problem formulation, not imported convergence guarantees.
Giacomini and Kitagawa	Sections 2.1–2.6 and Theorem 1 in CWP12/20	Prior sensitivity and classes of posterior beliefs.
Hoeting et al.	Equations (1)–(3); prior and predictive-evaluation discussion	Optional model-weighted predictions.
Merigoux et al.	Sections 3–4, default calculus and translation figures	Scopes, exceptions, empty/conflict outcomes.
Peyton Jones et al.	Sections 3–5, combinators and valuation semantics	Contract composition separate from valuation assumptions.
FinQA / TAT-QA	Dataset operations, annotation, model and evaluation sections; FinQA evaluation source	Evidence-linked local arithmetic; benchmark boundaries.
DocFinQA / FinBen	Context/retrieval and financial-task evaluations	Long-document difficulty and task coverage.
PubTables / GTE	Canonicalisation, dataset construction, methods and evaluation	Table structure and financial-domain extraction.
Cosentino et al.	Sections 3.1–3.2, 4.6 and 5	Persistent RAG provenance; scope of auditability evidence.

B.2 Version and access qualifications

Several discrepancies mattered. Doshi's thesis is dated 2024 despite the original filename. The archived Mernik text is the 2003 CWI report underlying the 2005 article. Giacomini and Kitagawa's copy is the April 2020 working paper underlying the 2021 article. The Docling v5 copy lists Auer as first author. The probabilistic-programming tutorial is an arXiv version revised in 2021, first submitted in 2018; a merged index DOI was not used. The Libkin author PDF has a footer DOI that resolves to another paper; title/author metadata from the indexes resolves the cited work to DOI `10.1145/2594538.2594561`. These discrepancies remain documented rather than being hidden by a cleaned bibliography.

McCarthy (1982), Imieliński and Lipski (1984), Tamer (2010), and Caiani and co-authors (2016) are retained as important blocked candidates. Access failures and rejected wrong-content downloads are recorded. The design does not use their uninspected technical claims. More specialised work on accounting measurement, process mining, multi-company equilibrium, actuarial inference and learned forecasting is recorded as expansion work for the corresponding implementation stage. The survey supports the core architecture without pretending that those later modules have a complete literature or rule base.

OpenAlex retraction flags and available Crossref correction/version relations were inspected

alongside local versions. No reviewed source was excluded for a discovered retraction. Missing correction metadata is marked unavailable; absence of a discovered notice does not certify that no notice exists. Citation counts are dated index metadata, and venue rankings were not available in this campaign. Neither is used as evidence of correctness.

B.3 Companion records and status

The six research ledgers are `source-support.json`, `citation-venue.json`, `backward-snowball.json`, `forward-snowball.json`, `claim-support.json`, and `omissions.json`. They record technical anchors, metadata sources, considered leads, omitted or blocked candidates and the distinction between source claims and this document's derivations. Citation leads with checked seed references are distinguished from cross-survey additions and topic expansion leads. Forward searches retain the most-cited and most-recent results returned by the recorded bounded queries; they do not establish exhaustive citation coverage. The separate `tools.json` records software evidence. Raw retrievals and text extracts permit a later reviewer to revisit the decisions.

The repair adds Mack's reserving paper, with its conditional-mean/variance assumptions, proofs, aggregate-error dependence and numerical comparisons inspected. It also archives official AASB texts incorporating IAS 2, IAS 32 and IFRS 17, distinguishing their Australian additions and applicable editions. The corporate filings are evidence of reported amounts and policies, rather than substitutes for every underlying standard. The ASBE 1 reproduction has not been verified against an accessible official current text, and official FASB/ASC access remains incomplete. These limitations are recorded at the specific operations they affect.

Every one of the 179 unique candidates in the acquired forward-query files now has an individual title/metadata relevance decision. This includes previously omitted discovery files associated with inaccessible seed papers. The generation script requires an explicit decision and stops on an unseen candidate. Screening is still distinct from technical reading: deferred provenance, temporal-query, bank-optimisation, extraction and actuarial follow-ups remain identified expansion risks. An omitted-paper reason does not make that paper technically reviewed or make the survey exhaustive.

This document specifies the product; it does not report a completed engine. The companion review records cover deterministic example arithmetic, bibliography/build checks, rendered-page inspection and author review. Human comprehension review, independent accounting review and implementation conformance remain separate future evidence. They are not replaced by a clean LaTeX build or by the quantity of literature collected.

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